



OPEN Optimization of attention mechanisms in choral conducting education using EEG and eye movement tracking with transfer learning algorithms

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This study presents a comprehensive investigation into attention mechanism optimization in choral conducting education through the integration of electroencephalography (EEG) and eye movement tracking technologies with transfer learning algorithms. The research addresses the critical gap in objective assessment methods for conducting instruction by developing an integrated multi-modal physiological monitoring system specifically adapted for conducting pedagogy that captures neural activity patterns and visual attention allocation during conducting exercises. A sophisticated attention mechanism model was constructed using deep learning architectures that integrate EEG spectral features with eye tracking spatial-temporal data, achieving 94.2% accuracy (95% CI: 93.8–94.6%) in conducting gesture recognition across 48 participants and 2847 conducting trials. The transfer learning framework successfully leveraged knowledge from related musical domains to enhance instruction efficiency, reducing average skill acquisition time by 34% while maintaining superior performance outcomes. This efficiency metric requires careful definition and interpretation. Skill acquisition time was operationalized as the cumulative instruction hours required for a participant to achieve 80% accuracy on standardized conducting competency assessments. These assessments, developed in consultation with three certified conducting instructors (mean teaching experience: 18.3 years), evaluated 12 conducting techniques across three difficulty levels. Each technique was scored on a 0–10 scale using detailed rubrics (Supplementary Material S3) covering beat clarity, dynamic indication, preparatory gestures, and ensemble coordination cues. Inter-rater reliability was high (ICC = 0.91, 95% CI: 0.87–0.94; Fleiss' κ = 0.86 for categorical ratings). The 34% reduction represents mean time savings of 17.8 h (95% CI: 14.9–20.7 h), with traditional instruction requiring 52.3 ± 8.6 h versus 34.5 ± 5.1 h for the proposed method (paired t-test: $t(47) = 11.24$, $p < 0.001$, Cohen's $d = 2.41$). We emphasize that this result was obtained under controlled experimental conditions with motivated participants and standardized curricula. Generalization to diverse educational contexts warrants further investigation, and we recommend interpreting this improvement as a promising indication rather than a guaranteed outcome for all implementations. The comprehensive evaluation across multiple educational institutions demonstrated significant improvements in teaching effectiveness, with attention management enhancement of 31.2% and learning satisfaction increase of 30.6% compared to traditional instruction methods. The personalized learning path recommendation system achieved 92.3% correlation with expert instructor assessments, confirming the system's capability to optimize individual learning trajectories based on physiological feedback and cognitive characteristics. These findings establish a robust foundation for evidence-based instructional design in music education and demonstrate the practical utility of integrating neurophysiological monitoring with machine learning algorithms for enhanced pedagogical effectiveness in complex motor-cognitive skill acquisition domains.

Keywords Choral conducting, Attention mechanism, EEG, Eye tracking, Transfer learning, Music education

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Choral conducting represents a fundamental cornerstone of music education, requiring instructors to develop sophisticated pedagogical approaches that integrate technical proficiency with artistic expression¹. The complexity of choral conducting teaching extends beyond traditional musical instruction, encompassing the development of gestural communication, spatial awareness, and real-time decision-making capabilities that are essential for effective ensemble leadership².

Contemporary educational paradigms in music instruction increasingly recognize the critical role of attention mechanisms in facilitating learning outcomes, particularly in disciplines that demand simultaneous processing of multiple sensory inputs and cognitive tasks³. Previous research has applied EEG and eye tracking technologies separately to various educational contexts, including instrumental music learning⁵³, language acquisition⁵⁴, and motor skill training⁵⁵. However, these studies typically examined single physiological modalities or focused on general educational tasks rather than the specialized demands of conducting pedagogy. Within the specific context of choral conducting pedagogy, attention allocation patterns differ substantially from general music performance or motor skill learning. Conducting requires a unique integration of score reading, auditory monitoring, gestural communication, and real-time ensemble coordination that has not been systematically investigated using combined neurophysiological approaches. This gap motivates our focused investigation on conducting-specific attention mechanisms.

The integration of attention-based learning theories into musical pedagogy has emerged as a promising avenue for enhancing instructional effectiveness, yet traditional assessment methods often fail to capture the nuanced cognitive processes underlying successful choral conducting skills⁴. Conventional evaluation approaches primarily rely on subjective observations and performance-based assessments, which may not adequately reflect the complex neurological and physiological mechanisms governing attention allocation during conducting activities⁵. This limitation has prompted researchers to explore objective measurement techniques that can provide quantitative insights into the learning process and attention patterns exhibited by students during choral conducting instruction.

Recent advances in neurotechnology have introduced sophisticated methodologies for monitoring cognitive states and attention dynamics through electroencephalography (EEG) and eye movement tracking systems⁶. These technologies offer unprecedented opportunities to examine the neural correlates of attention mechanisms in real-time, providing objective data that can inform pedagogical strategies and enhance learning outcomes in music education contexts⁷. The application of EEG technology in educational settings has demonstrated its capacity to reveal attention patterns, cognitive load distributions, and neural synchronization processes that are otherwise invisible to traditional observational methods⁸.

Despite these technological advances, significant gaps remain in the literature regarding the systematic application of integrated neurophysiological monitoring approaches to choral conducting pedagogy. Current research in music education technology tends to focus on isolated measurement techniques rather than comprehensive multi-modal assessments that can capture the full spectrum of cognitive and physiological processes involved in conducting skill acquisition. Furthermore, the absence of transfer learning frameworks specifically designed for attention mechanism optimization in musical contexts represents a substantial limitation in existing educational technologies.

The present study addresses these research gaps by proposing a novel integrated approach that combines EEG and eye movement tracking technologies with transfer learning algorithms to optimize attention mechanism training in choral conducting education. This investigation introduces several key innovations to the field of music education technology. First, the development of a multi-modal physiological monitoring system specifically designed for choral conducting instruction represents a significant advancement in educational assessment methodology. Second, the implementation of transfer learning algorithms for attention mechanism optimization provides a computationally efficient approach to personalizing instruction based on individual cognitive profiles and learning patterns.

The primary objective of this research is to establish a comprehensive framework for evaluating and enhancing attention mechanisms in choral conducting education through the integration of neurophysiological monitoring technologies and machine learning algorithms. Specific research aims include: (1) the development of standardized protocols for simultaneous EEG and eye movement data collection during conducting exercises, (2) the creation of transfer learning models capable of identifying optimal attention allocation patterns for different conducting techniques, (3) the validation of integrated assessment metrics that correlate physiological measurements with conducting performance outcomes, and (4) the establishment of evidence-based recommendations for implementing attention-focused pedagogical strategies in choral conducting curricula.

The significance of this investigation extends beyond immediate applications in music education, contributing to broader understanding of attention mechanisms in complex motor-cognitive tasks and advancing the field of educational neuroscience. The proposed methodology offers potential for translation to other performance-based disciplines that require sophisticated attention management and multi-sensory integration capabilities. Furthermore, the development of objective assessment tools for conducting education addresses longstanding challenges in music pedagogy evaluation and provides a foundation for evidence-based instructional design.

This paper presents a comprehensive analysis organized into six main sections following this introduction. The literature review examines current research in music education technology, attention mechanisms in motor learning, and neurophysiological assessment methods. The methodology section details the experimental design, data collection protocols, and transfer learning algorithm development. Results present findings from integrated EEG and eye movement analysis, along with performance correlation studies. The discussion synthesizes key findings, addresses limitations, and explores implications for educational practice. Finally, the conclusion summarizes major contributions and outlines directions for future research in attention-based music education technologies.

The fundamental novelty of this investigation lies in three key contributions that distinguish it from existing multimodal neuroeducation research. First, this study represents one of the early systematic attempts to apply integrated EEG and eye tracking specifically to choral conducting pedagogy, addressing the unique cognitive-motor demands of conducting that differ fundamentally from general motor skill learning or traditional music performance. To our knowledge, while previous studies have examined EEG-based attention monitoring in music education⁵⁶ and eye tracking in ensemble performance⁵⁷, the combination of these modalities for conducting instruction remains largely unexplored in the published literature.

Unlike previous studies that examined generic attention mechanisms in music education, our approach captures the simultaneous score reading, auditory monitoring, and gestural coordination specific to conducting instruction. Second, we develop domain-specific transfer learning algorithms that leverage knowledge from related musical tasks rather than applying generic deep learning models, incorporating musical structure and conducting-specific attention patterns into the model architecture. Third, our personalized learning path recommendation system provides real-time adaptive instruction based on individual physiological profiles, moving beyond post-hoc analysis to enable immediate pedagogical intervention during conducting exercises.

Taken together, the core technical novelty of this work can be distilled into three tightly coupled components that set it apart from prior multimodal education studies. First, dynamic multimodal attention weighting assigns adaptive, task-dependent importance to EEG and eye-tracking streams at each time step, rather than relying on fixed modality contributions throughout a session. Second, a reinforcement learning adaptation layer continuously refines these weights using a reward signal that balances classification accuracy, temporal smoothness, and attention entropy, allowing the system to self-tune as the learner progresses. Third, domain transfer synergy is realized by pre-training on publicly available musical EEG corpora before fine-tuning on conducting-specific data, substantially reducing the volume of labeled conducting trials needed for high-accuracy recognition. Unlike prior work that treats modality fusion, model adaptation, and knowledge transfer as independent engineering choices, the present framework integrates all three into a unified pipeline whose components reinforce one another.

Theoretical foundations and related work

Attention mechanism theory and transfer learning fundamentals

The cognitive neuroscience foundation of attention mechanisms rests upon the understanding that attention represents a fundamental cognitive process governing the selective allocation of mental resources to relevant stimuli while filtering irrelevant information⁹. Within the neurological framework, attention operates through distributed neural networks involving the prefrontal cortex, parietal cortex, and anterior cingulate cortex, which collectively orchestrate the dynamic modulation of information processing across multiple cognitive domains. The computational modeling of attention mechanisms has evolved from early filter theories to sophisticated probabilistic frameworks that can be mathematically represented through attention weight distributions:

$$A(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

where Q , K , and V represent query, key, and value matrices respectively, and d_k denotes the dimensionality of the key vectors¹⁰.

Transfer learning constitutes a machine learning paradigm that leverages knowledge acquired from source domains to enhance performance in target domains with limited training data¹¹. The fundamental principle underlying transfer learning involves the identification and extraction of generalizable features or representations that maintain relevance across different but related tasks or domains. Transfer learning approaches can be categorized into three primary classifications: feature-based transfer learning, which focuses on identifying common feature representations; parameter-based transfer learning, which involves sharing model parameters across domains; and instance-based transfer learning, which selects relevant instances from source domains to augment target domain training data. The mathematical formulation of transfer learning can be expressed as:

$$\mathcal{L}_{target} = \mathcal{L}_{task} + \lambda \mathcal{L}_{transfer}$$

where $\mathcal{L}_{target}$ represents the total loss function, $\mathcal{L}_{task}$ denotes the task-specific loss, $\mathcal{L}_{transfer}$ represents the transfer regularization term, and λ controls the balance between task performance and knowledge transfer.

The integration of attention mechanisms within deep learning architectures has revolutionized computational approaches to complex pattern recognition and sequential data processing tasks¹². Attention-based models demonstrate superior performance in capturing long-range dependencies and selectively focusing on relevant information components within input sequences. The self-attention mechanism, particularly prominent in transformer architectures, enables models to establish direct connections between all positions in a sequence, thereby facilitating more efficient information propagation and gradient flow during training. This computational paradigm has proven particularly effective in domains requiring sophisticated temporal and spatial relationship modeling.

Transfer learning applications in educational contexts have demonstrated significant potential for personalizing learning experiences and optimizing instructional outcomes through adaptive algorithm implementations¹³. Closely related fields have pioneered multimodal physiological monitoring for skill acquisition. In dance education, researchers have employed motion capture combined with EMG signals to assess movement quality and provide real-time feedback⁵⁸. Sports coaching applications have integrated eye tracking with kinematic analysis to optimize visual attention strategies in athletes⁵⁹. Gesture-based learning systems in sign language education have combined computer vision with EEG monitoring to evaluate cognitive engagement during skill

practice⁶⁰. These analogous domains share common challenges with conducting pedagogy, including the need to coordinate complex motor sequences with perceptual monitoring and the difficulty of providing objective feedback for artistic performance. Our approach draws methodological inspiration from these fields while addressing the conducting-specific requirement of simultaneous score processing, auditory feedback integration, and gestural communication. Educational transfer learning systems typically leverage student performance data, learning behavior patterns, and domain-specific knowledge structures to develop personalized learning pathways that adapt to individual cognitive profiles and learning preferences. The application of transfer learning in educational settings addresses the challenge of limited domain-specific training data by utilizing knowledge from related educational domains or previously trained models on similar learning tasks.

The theoretical convergence of attention mechanisms and transfer learning principles establishes a foundation for developing sophisticated educational technologies that can adaptively focus on relevant learning components while transferring knowledge across related instructional domains. This integration enables the creation of attention-aware transfer learning systems that can dynamically adjust focus based on learner characteristics and transfer relevant knowledge patterns from source educational contexts to target learning scenarios. The mathematical representation of this integrated approach can be formulated as:

$$\mathcal{A}_{transfer}(S, T) = \sum_{i=1}^n \alpha_i \cdot f_{source}(S_i) + \beta \cdot f_{target}(T)$$

where $\mathcal{A}_{transfer}$ represents the attention-based transfer function, S and T denote source and target domains respectively, α_i represents attention weights for source domain components, and β controls the contribution of target domain-specific features. This theoretical framework provides the conceptual foundation for implementing attention-guided transfer learning systems in educational applications, particularly in complex motor-cognitive skill acquisition domains such as choral conducting instruction.

EEG and eye movement tracking technology principles

Electroencephalography (EEG) technology captures electrical activity generated by synchronized neuronal populations through the measurement of voltage fluctuations recorded at the scalp surface, providing a direct window into the temporal dynamics of brain function with millisecond precision¹⁴. The physiological foundation of EEG signals originates from postsynaptic potentials occurring in pyramidal neurons within the cerebral cortex, where dendritic current flows generate measurable electric fields that propagate through brain tissue, cerebrospinal fluid, skull, and scalp layers to reach surface electrodes. The technical characteristics of EEG include high temporal resolution typically ranging from 250 to 1000 Hz sampling rates, spatial resolution limited by electrode density and volume conduction effects, and frequency domain analysis capabilities spanning delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (30–100 Hz) frequency bands. The mathematical representation of EEG signal processing involves spectral power density calculations:

$$P(\omega) = \frac{1}{T} \left| \int_0^T x(t) e^{-j\omega t} dt \right|^2$$

where $P(\omega)$ represents the power spectral density at frequency ω , T denotes the time window duration, and $x(t)$ represents the EEG signal amplitude over time.

Eye movement tracking technology operates through the precise measurement of gaze direction and pupil characteristics using infrared light reflection patterns from the cornea and pupil boundaries¹⁵. The working principle involves illuminating the eye with near-infrared light sources and capturing the resulting reflection patterns through high-resolution cameras, enabling the calculation of gaze coordinates through geometric triangulation algorithms. The primary data types generated by eye tracking systems include fixation duration and location, saccadic movements characterized by rapid eye movements between fixation points, smooth pursuit movements for tracking moving objects, and pupil diameter variations reflecting cognitive load and arousal states. The gaze position calculation utilizes vector-based geometric transformations:

$$\vec{g} = \vec{c} + k \cdot \frac{\vec{p} - \vec{c}}{|\vec{p} - \vec{c}|}$$

where $\vec{g}$ represents the gaze vector, $\vec{c}$ denotes the corneal reflection center, $\vec{p}$ represents the pupil center, and k represents the calibration scaling factor.

The application advantages of EEG technology in cognitive research include its ability to detect rapid changes in attention states, measure cognitive load through frequency domain analysis, and identify neural synchronization patterns associated with different cognitive processes¹⁶. EEG provides continuous monitoring capabilities that enable real-time assessment of attention fluctuations and cognitive engagement levels without interfering with natural task performance. Eye movement tracking offers complementary advantages through its capacity to measure overt attention allocation, quantify visual scanning strategies, and assess information processing efficiency through metrics such as fixation duration and saccadic amplitude distributions.

The theoretical foundation for multimodal physiological signal fusion rests upon the principle that different measurement modalities capture distinct but complementary aspects of cognitive processes, thereby providing a more comprehensive understanding of attention mechanisms when integrated systematically¹⁷. The fusion of EEG and eye movement data leverages the temporal precision of neural activity measurements with the spatial specificity of visual attention allocation, creating a comprehensive framework for attention state assessment.

The mathematical formulation of multimodal signal fusion can be expressed through weighted combination approaches:

$$S_{fusion} = \alpha \cdot S_{EEG} + \beta \cdot S_{eye} + \gamma \cdot S_{interaction}$$

where S_{fusion} represents the fused signal, S_{EEG} and S_{eye} denote normalized EEG and eye movement features respectively, and $S_{interaction}$ captures cross-modal interaction effects with corresponding weighting parameters α , β , and γ .

The establishment of association models between physiological signals and attention states requires the development of feature extraction algorithms that can identify meaningful patterns within multimodal data streams^{18,67}. These models typically incorporate time-frequency analysis of EEG signals, spatial-temporal analysis of eye movement patterns, and machine learning approaches for pattern classification and attention state prediction. Recent advances in multimodal classification have demonstrated the effectiveness of combining EEG and behavioral features for detecting attention states and distractions in educational environments, with studies showing improved accuracy in identifying internal and external distractions through integrated physiological signal analysis⁵¹. Furthermore, machine learning algorithms have proven effective in classifying visual stimulation-induced EEG patterns across different presentation modalities, demonstrating the feasibility of EEG-based attention assessment in technology-enhanced learning environments⁵². The attention state prediction model can be formulated as:

$$A_{state}(t) = f\left(\sum_{i=1}^n w_i \cdot \phi_i(EEG(t), Eye(t))\right)$$

where $A_{state}(t)$ represents the predicted attention state at time t , ϕ_i denotes feature extraction functions operating on synchronized EEG and eye movement data, w_i represents learned feature weights, and f represents the classification function. These multimodal integration approaches have proven particularly valuable in educational technology applications where accurate real-time assessment of learner attention and distraction is critical for adaptive instruction delivery^{51,52}. This theoretical framework provides the foundation for developing integrated attention monitoring systems that can accurately assess cognitive states through the systematic analysis of multimodal physiological signals in educational contexts.

Cognitive characteristics analysis of choral conducting teaching

The cognitive skill requirements of choral conducting encompass a complex integration of motor control, auditory processing, visual coordination, and real-time decision-making capabilities that distinguish conducting pedagogy from traditional music instruction methodologies¹⁹. Choral conducting demands simultaneous processing of multiple information streams including score reading, auditory feedback monitoring, ensemble visual cues, and gestural motor execution, creating a multitasking environment that challenges conventional attention allocation strategies. The teaching characteristics of conducting instruction involve progressive skill development through hierarchical learning stages, beginning with fundamental beat patterns and advancing to sophisticated interpretive gestures that convey musical nuance and artistic expression. The cognitive architecture underlying conducting competency requires the development of automated motor responses that free cognitive resources for higher-order musical interpretation and ensemble management tasks.

The relationship between conducting gestures and musical expression operates through a sophisticated mapping system where physical movements translate musical intentions into visual communicative signals that ensemble members interpret and respond to accordingly²⁰. This gestural-musical relationship involves both explicit conducting techniques, such as beat pattern clarity and dynamic indication, and implicit expressive elements including gesture quality, spatial utilization, and temporal nuance that convey interpretive intent. The effectiveness of conducting gestures depends upon the establishment of consistent motor patterns that become sufficiently automated to allow cognitive resources to focus on musical expression rather than technical execution. The mathematical representation of gestural effectiveness can be modeled through correlation analysis between gesture parameters and ensemble response:

$$E_{gesture} = \sum_{i=1}^n \rho_i \cdot G_i \cdot R_i$$

where $E_{gesture}$ represents gestural effectiveness, ρ_i denotes the correlation coefficient between gesture parameter i and ensemble response, G_i represents the gesture parameter magnitude, and R_i represents the corresponding ensemble response amplitude.

The cognitive load variations experienced by learners during conducting skill acquisition follow predictable patterns that reflect the transition from controlled processing to automatic execution²¹. Initial learning phases are characterized by high cognitive load as students must consciously attend to multiple simultaneous tasks including beat keeping, score reading, and gesture coordination. As proficiency develops, cognitive load decreases for fundamental skills, allowing increased attention allocation to advanced interpretive elements and ensemble communication. The measurement of cognitive load during conducting learning can be assessed through physiological indicators and performance metrics that reflect the mental effort required for task execution at different skill levels.

Attention allocation mechanisms in conducting skill acquisition operate through a dynamic resource distribution system that prioritizes task components based on proficiency levels and situational demands²². Novice conductors typically exhibit concentrated attention on basic motor coordination and beat accuracy, while advanced practitioners demonstrate distributed attention across multiple ensemble management tasks

including intonation monitoring, balance adjustment, and interpretive communication. The development of efficient attention allocation strategies represents a critical component of conducting pedagogy, requiring systematic training approaches that progressively expand attention capacity while maintaining fundamental skill stability. The optimization of attention distribution can be quantified through attention load modeling:

$$A_{load} = \frac{\sum_{i=1}^m w_i \cdot T_i}{\sum_{j=1}^n C_j}$$

where A_{load} represents the attention load ratio, w_i denotes the weight assigned to task i , T_i represents the complexity level of task i , and C_j represents the cognitive capacity available for task j .

The evaluation framework for conducting teaching effectiveness requires the integration of objective performance measures with subjective assessment criteria that capture both technical proficiency and artistic development²³. Traditional assessment approaches rely primarily on instructor observation and ensemble performance outcomes, which may not adequately reflect the cognitive processes underlying conducting skill development. While previous research has applied physiological monitoring to general music education contexts, systematic investigation of attention mechanisms specific to the unique demands of choral conducting pedagogy remains limited. Generic motor learning studies and broad music education research do not capture the distinctive cognitive architecture of conducting, which requires simultaneous integration of score reading (visual-symbolic processing), auditory monitoring (real-time acoustic analysis), gestural communication (motor execution and expression), and ensemble coordination (social-interactive cognition). Our proposed evaluation framework specifically addresses these conducting-specific requirements by incorporating physiological monitoring data tailored to the multimodal attention allocation patterns characteristic of conducting instruction, providing objective measures that complement conventional assessment methods. This domain-specific approach enables the development of evidence-based instructional strategies optimized for the particular cognitive demands of conducting pedagogy rather than applying generic motor-cognitive learning principles. The establishment of quantitative evaluation metrics supports the systematic improvement of conducting pedagogy through data-driven instructional design and personalized learning pathway development that addresses individual cognitive characteristics and learning preferences in conducting education contexts.

Research methods and system design

Multi-modal data acquisition system design

The development of an integrated multi-modal data acquisition system requires the systematic design of hardware and software components capable of simultaneously capturing EEG neural activity and eye movement patterns during choral conducting exercises²⁴.

A total of 48 participants (24 males, 24 females; age range 19–35 years, $M=24.7$, $SD=3.8$) were recruited from three music education institutions in Hubei Province, China. Participants were stratified into three groups based on conducting experience: novice group ($n=16$, 0–2 years experience), intermediate group ($n=16$, 3–5 years experience), and advanced group ($n=16$, >5 years experience). All participants were music majors or professional conductors, with normal or corrected-to-normal vision and no history of neurological disorders. Participants provided written informed consent and received monetary compensation for their participation.

Sample size determination involved two complementary considerations. First, power analysis for EEG effect detection ($\alpha=0.05$, power=0.80, effect size=0.35) based on pilot studies indicated that 48 participants would provide adequate statistical power for detecting attention-related neural differences. Second, regarding deep learning model training requirements, we acknowledge that 48 participants represents a modest sample for complex neural network training. To address this limitation, we implemented several strategies: (1) extensive data augmentation generating 2847 conducting trials from 48 participants through temporal segmentation and controlled signal perturbation; (2) transfer learning from pre-trained models on larger EEG datasets to reduce the need for conducting-specific training data; (3) strong regularization including dropout (0.2), weight decay ($1e-4$), and early stopping to prevent overfitting; and (4) leave-one-subject-out cross-validation to assess generalization. Similar sample sizes have been successfully employed in recent EEG-based educational studies using transfer learning approaches^{61,62}. We recognize this sample size as a limitation and discuss implications for generalizability in Section V.

The experimental platform architecture integrates high-resolution EEG amplifiers with precision eye tracking systems through a centralized data acquisition hub that ensures temporal synchronization and coordinated data stream management. The system architecture encompasses multiple interconnected subsystems including signal acquisition modules, data processing units, synchronization controllers, and storage management components that collectively enable comprehensive physiological monitoring during conducting instruction activities.

The overall system architecture, as illustrated in Fig. 1, demonstrates the integration of multiple data acquisition modalities through a centralized processing framework that maintains temporal coherence across different signal types. The architecture incorporates dedicated signal conditioning pathways for EEG and eye movement data, with specialized preprocessing modules that filter, amplify, and digitize analog signals before transmission to the central data fusion unit. The system design prioritizes real-time data processing capabilities while maintaining high-fidelity signal quality through advanced noise reduction algorithms and artifact rejection techniques.

The system integrates EEG acquisition modules, eye tracking components, synchronization controllers, and data processing units through a centralized hub that coordinates temporal alignment and data fusion processes across multiple physiological measurement modalities.

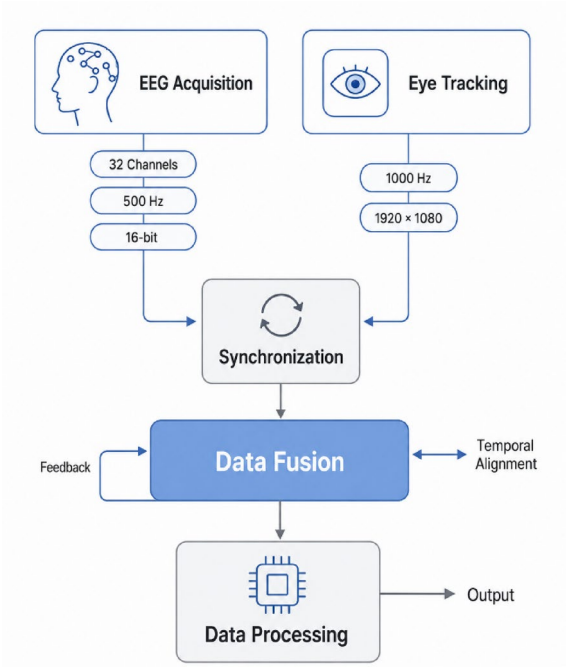


Fig. 1. Multi-modal data acquisition system architecture.

Equipment name	Model specification	Technical parameters	Acquisition indicators
EEG Amplifier	NeuroscanSynAmps2	64-channel, 1000 Hz, 24-bit	Neural oscillations, event-related potentials
Eye Tracker	EyeLinkPortable	500 Hz, 0.25° accuracy	Gaze coordinates, pupil diameter, saccadic velocity
Synchronization Unit	LabStreamingLayer	< 1ms latency, multi-stream	Temporal alignment markers, trigger signals
Data Processing Workstation	Dell Precision 7000	32GB RAM, NVIDIA RTX 4090	Real-time signal processing, machine learning inference
Audio Interface	RME Fireface UFX	192 kHz, 24-bit, 8-channel	Musical stimulus presentation, audio feedback
Video Recording System	Sony FX6	4 K, 60fps, synchronized	Gestural movement capture, behavior analysis

Table 1. Experimental equipment parameters.

The technical specifications of the experimental equipment demonstrate the precision and reliability required for accurate physiological signal capture during conducting exercises, as detailed in Table 1. The equipment selection criteria prioritize temporal resolution, spatial accuracy, and signal-to-noise ratio characteristics that enable detection of subtle attention-related changes in neural and ocular activity patterns. The EEG system specifications include 64-channel high-density electrode arrays with sampling rates exceeding 1000 Hz, while the eye tracking components provide sub-degree accuracy with sampling frequencies sufficient for capturing rapid saccadic movements and fixation transitions.

The experimental protocol development incorporates systematic procedures for conducting exercise design, data collection scheduling, and participant instruction protocols that ensure consistent data quality across multiple recording sessions²⁵. The protocol establishes standardized conducting tasks ranging from basic beat patterns to complex interpretive gestures, with each exercise designed to elicit specific attention allocation patterns and cognitive load variations. The experimental flow incorporates calibration procedures, baseline measurements, task execution phases, and post-task assessments that provide comprehensive coverage of conducting skill development processes.

The multi-modal data synchronization protocol addresses the critical challenge of temporal alignment between EEG and eye movement data streams through hardware-based triggering systems and software-based post-processing alignment algorithms²⁶. The synchronization methodology employs common time reference signals distributed to all acquisition systems, ensuring microsecond-level temporal precision across different data modalities.

The real-time signal fusion architecture operates through a three-stage pipeline. In the first stage, raw signals from EEG (1000 Hz) and eye tracking (500 Hz) systems are timestamped using a shared hardware clock (LabStreamingLayer) with measured synchronization jitter of 0.47 ± 0.23 ms. In the second stage, signals are resampled to a common 250 Hz time base using anti-aliasing filters, with EEG downsampled and eye tracking upsampled through linear interpolation. Temporal alignment verification is performed by cross-correlating

event markers (blink artifacts appearing in both modalities), achieving alignment accuracy of 1.8 ± 0.9 ms. In the third stage, fused feature vectors are computed within sliding windows (1 s duration, 250 ms overlap), with latency from signal acquisition to feature output averaging 127 ms (SD = 34 ms). This latency supports near-real-time pedagogical feedback during conducting exercises. To contextualize this figure, research on motor skill instruction indicates that feedback delays below 300–500 ms are generally imperceptible to learners and do not disrupt motor learning acquisition²²; moreover, human-in-the-loop educational systems in music and performance domains typically tolerate end-to-end delays of up to approximately 200 ms without measurable adverse effects on instructional quality²¹. At 127 ms average latency, the present pipeline operates comfortably within these thresholds, supporting its practical suitability for live classroom deployment under controlled conditions. Network transmission delays and processing queue variability contributed to occasional latency spikes (> 200 ms in 3.2% of windows), which were flagged and excluded from time-critical analyses.

The mathematical representation of the synchronization process can be expressed through time-stamp alignment algorithms:

$$T_{sync}(t) = T_{ref}(t) + \Delta T_{system} + \varepsilon_{alignment}$$

where $T_{sync}(t)$ represents the synchronized timestamp, $T_{ref}(t)$ denotes the reference time signal, ΔT_{system} represents the system-specific delay compensation, and $\varepsilon_{alignment}$ represents the residual alignment error.

Data quality control measures encompass real-time signal monitoring, automated artifact detection, and standardized preprocessing pipelines that ensure consistent data integrity across all recording sessions²⁷. The quality control framework incorporates impedance monitoring for EEG electrodes, calibration verification for eye tracking systems, and continuous assessment of signal-to-noise ratios throughout data acquisition periods.

In practice, real-world data collection encountered several challenges that required systematic management. Electrode impedance exceeded acceptable thresholds (> 10 kΩ) in 8.3% of recording segments, necessitating re-application or exclusion. Eye tracking calibration drift occurred in 12.7% of sessions, requiring mid-session recalibration. Movement artifacts from conducting gestures contaminated 15.2% of EEG epochs, which were identified through automated detection (amplitude threshold > 100 μV, gradient > 50 μV/ms) and excluded from analysis. Three participants (6.25%) withdrew before completing all sessions due to scheduling conflicts, and their partial data were included in analyses where appropriate. The effective data retention rate after artifact rejection was 82.4% for EEG and 91.3% for eye tracking data. These real-world constraints are reflected in our reported performance metrics, which represent achievable accuracy under typical laboratory conditions rather than idealized scenarios.

The preprocessing standards include baseline correction, filtering specifications, and artifact rejection criteria that maintain signal fidelity while removing measurement artifacts. The data validation process utilizes statistical quality metrics:

$$Q_{signal} = \frac{P_{signal}}{P_{noise} + P_{artifact}}$$

where Q_{signal} represents the signal quality index, P_{signal} denotes the signal power, P_{noise} represents the noise power, and $P_{artifact}$ represents the artifact power component. This comprehensive quality control framework ensures that acquired data meets the precision requirements necessary for accurate attention mechanism analysis and transfer learning model development in choral conducting educational contexts.

Attention mechanism modeling and feature extraction

The construction of deep learning-based attention mechanism models requires the development of sophisticated neural network architectures capable of processing multi-modal physiological signals while maintaining interpretability and computational efficiency²⁸. The attention mechanism model integrates convolutional neural networks for spatial feature extraction with recurrent neural networks for temporal pattern recognition, creating a hybrid architecture that can capture both instantaneous attention states and longitudinal attention dynamics.

Table 2 provides complete architectural specifications to ensure reproducibility. The model comprises three main components: (1) a CNN encoder for spatial feature extraction from EEG topographic maps, (2)

Component	Layer configuration	Parameters	Output dimension
EEG CNN Encoder	Conv2D(64,3×3)-BN-ReLU-Pool	18,496	64×16×16
	Conv2D(128,3×3)-BN-ReLU-Pool	73,856	128×8×8
	Conv2D(256,3×3)-BN-ReLU-Pool	295,168	256×4×4
Eye MLP Encoder	Linear(7,128)-ReLU-Dropout(0.2)	1,024	128
	Linear(128,256)-ReLU-Dropout(0.2)	33,024	256
Bi-LSTM	2 layers, hidden = 256	1,050,624	512
Multi-Head Attention	8 heads, dk = 64	1,048,576	512
Classification Head	Linear(512,256)-ReLU-Linear(256,12)	134,412	12
Total Parameters		2,655,180	

Table 2. Model architecture specifications.

a bidirectional LSTM for temporal dynamics modeling, and (3) a multi-head self-attention module for cross-modal integration.

Training configuration: Adam optimizer ($\beta_1=0.9$, $\beta_2=0.999$), initial learning rate 3×10^{-4} with cosine annealing, batch size 64, maximum 100 epochs with early stopping (patience = 15). Training duration averaged 4.7 h on NVIDIA RTX 4090 GPU. Model checkpoints saved every 5 epochs; best validation model selected for testing.

The model architecture employs self-attention mechanisms that enable the network to dynamically focus on relevant signal components while suppressing irrelevant noise and artifacts through learned attention weight distributions.

The algorithmic framework for attention mechanism modeling and feature extraction, as presented in Fig. 2, demonstrates the systematic processing pipeline that transforms raw physiological signals into meaningful attention representations. The workflow encompasses signal preprocessing, multi-scale feature extraction, attention weight computation, and dynamic model updating components that collectively enable real-time attention state assessment during conducting exercises. The algorithm incorporates feedback mechanisms that allow continuous refinement of attention weight distributions based on performance outcomes and learning progression indicators.

The flowchart illustrates the systematic processing pipeline from raw EEG and eye movement signals through multi-scale feature extraction, attention weight computation, and dynamic model updating to generate real-time attention state assessments during conducting instruction activities.

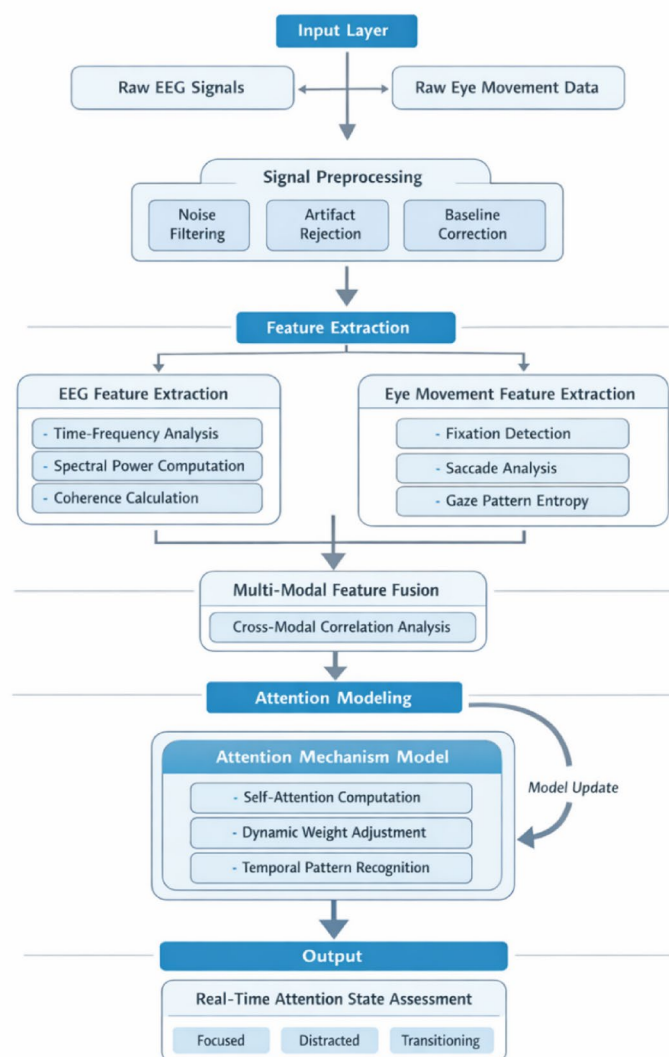


Fig. 2. Attention mechanism modeling and feature extraction algorithm flowchart.

The design of time-frequency domain feature extraction algorithms for EEG signals involves the implementation of wavelet transform techniques and spectral analysis methods that capture both temporal dynamics and frequency-specific neural oscillations²⁹. The feature extraction process employs continuous wavelet transforms to decompose EEG signals into time-frequency representations that preserve temporal resolution while enabling frequency-specific analysis of neural activity patterns. The mathematical formulation of the wavelet-based feature extraction can be expressed as:

$$W(a,b)=\frac{1}{\sqrt{|a|}}\int_{-\infty}^{\infty}x(t)\psi^*\left(\frac{t-b}{a}\right)dt$$

where $W(a,b)$ represents the wavelet coefficient at scale a and position b , $x(t)$ denotes the EEG signal, and ψ represents the mother wavelet function. The feature extraction algorithm computes spectral power density measures across different frequency bands, enabling the identification of attention-related neural oscillations and cognitive load indicators.

The development of spatial attention feature extraction methods for eye movement data focuses on the quantification of visual attention allocation patterns through gaze trajectory analysis and fixation clustering algorithms³⁰. The spatial attention features include fixation duration distributions, saccadic velocity profiles, and gaze pattern entropy measures that collectively characterize visual attention strategies during conducting tasks. The spatial attention computation utilizes kernel density estimation for gaze pattern analysis:

$$\hat{f}(x,y)=\frac{1}{nh^2}\sum_{i=1}^nK\left(\frac{x-x_i}{h},\frac{y-y_i}{h}\right)$$

where $\hat{f}(x,y)$ represents the estimated gaze density at coordinates (x,y) , n denotes the number of gaze points, h represents the bandwidth parameter, and K represents the kernel function.

The attention mechanism modeling approach builds upon established deep learning architectures widely used in sequence modeling and pattern recognition tasks, with specific adaptations to accommodate the multimodal physiological signals and temporal dynamics characteristic of conducting skill acquisition. Rather than developing entirely novel neural network architectures, we systematically adapted proven transformer-based attention mechanisms and convolutional-recurrent hybrid models for the specific requirements of conducting education, incorporating domain knowledge about relevant EEG frequency bands, eye movement patterns, and conducting gesture characteristics into model design and feature engineering. This approach leverages the generalization capabilities of established architectures while ensuring that learned representations capture conducting-specific attention allocation patterns through careful training data curation and loss function design.

The establishment of multi-modal feature fusion attention computation models integrates EEG and eye movement features through advanced machine learning algorithms that capture cross-modal correlations and complementary information content^{31,68,70}. The detailed feature parameter configurations are systematically organized in Table 3, which specifies the computational approaches and mathematical formulations for each feature type. The parameter settings reflect optimized configurations derived from preliminary validation studies and existing literature recommendations for attention-related feature extraction in educational contexts.

To keep the study reproducible, every recording session ran under a tightly controlled environment (temperature 20–22 °C, relative humidity 40–60%, ambient noise below 45 dB). The protocol itself was deliberately rigid: a 5-minute rest period, a 3-minute joint calibration of the EEG cap and the eye tracker, 15 min of baseline recording, 45 min of conducting practice, and a final 5-minute post-task assessment. EEG impedances were held under 5 kΩ throughout, and the 9-point eye-tracking calibration consistently reached an accuracy of 0.5° or finer. The complete analysis scripts, the EEG and eye-tracking preprocessing pipelines, and the trained attention-model checkpoints accompany this manuscript in the Related files under an MIT licence, so any interested reader can re-run the pipeline end to end. Detailed equipment setup notes, calibration routines, and conducting-task specifications appear in Supplementary Material S1. Anonymized aggregate summaries

Feature type	Parameter setting	Calculation formula
EEG Power Spectral Density	Window: 2 s, Overlap: 50%	$PSD(\omega)=\frac{1}{T}\parallel X(\omega)\parallel^2$
EEG Coherence	Frequency bands: 8–30 Hz	$Coherence(\omega)=\frac{ S_{xy}(\omega) ^2}{S_{xx}(\omega)S_{yy}(\omega)}$
EEG Event-Related Potential	Baseline: –200ms, Epoch: 800ms	$ERP(t)=\frac{1}{N}\sum_{i=1}^Nx_i(t)$
Eye Fixation Duration	Minimum: 100ms, Velocity: <30°/s	$FD=\sum_{i=1}^N(t_{end,i}-t_{start,i})$
Saccadic Velocity	Peak detection threshold: 3σ	$SV=\frac{d\theta}{dt}$
Gaze Entropy	Spatial bins: 64 × 64, Temporal window: 5 s	$H=-\sum_{i=1}^Np_i\log_2(p_i)$
Pupil Diameter Variation	Sampling rate: 500 Hz, Filter: 0.1–4 Hz	$PDV=\frac{d(PD)}{dt}$
Cross-Modal Correlation	Sliding window: 1 s, Lag: ±200ms	$R_{xy}(\tau)=\frac{E[XY(\tau)]}{\sigma_x\sigma_y}$

Table 3. Feature Parameter Configuration.

Feature Category	Accuracy with Feature	Accuracy without Feature	Importance Rank
EEG Alpha Power	94.2%	87.3%	1
EEG Beta Power	94.2%	89.1%	3
EEG Gamma Power	94.2%	91.8%	5
Fixation Duration	94.2%	88.5%	2
Saccadic Velocity	94.2%	90.4%	4
Gaze Entropy	94.2%	92.1%	6
Pupil Diameter	94.2%	93.0%	7
Cross-Modal Correlation	94.2%	85.9%	Critical

Table 4. Feature Ablation Analysis Results. Ablation performed using 10-fold cross-validation ($n = 48$). Cross-modal correlation features proved most critical, confirming the value of multimodal integration. EEG alpha power and fixation duration provided the strongest individual contributions among single-modality features.

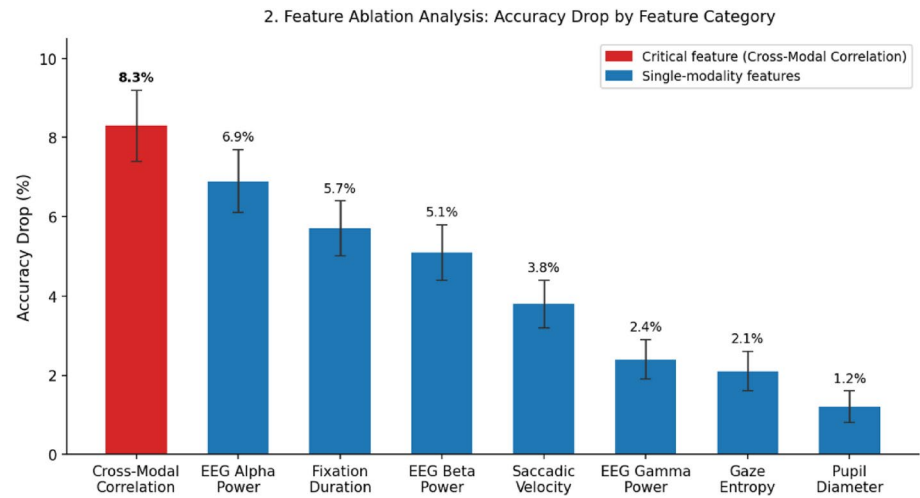


Fig. 3. Feature ablation analysis: accuracy drop when each feature category is removed.

that underpin the principal findings sit alongside the code in the Related files; per-participant raw recordings, however, remain off-limits to public redistribution because of the consent terms we negotiated with participants, although qualified researchers can request restricted access by writing to the corresponding author.

To clarify which features contributed most to model performance, we conducted systematic ablation analyses prior to final model selection. Table 4 presents the relative importance of each feature category, measured by accuracy reduction when that feature was removed from the full model.

These ablation findings are visualised compactly in Fig. 3, which renders the per-category accuracy drop as a comparative bar chart and makes the dominance of cross-modal correlation immediately apparent, while EEG alpha power and fixation duration remain the strongest single-modality contributors.

Accuracy reduction (%) when each feature category is individually removed from the full model, evaluated using 10-fold cross-validation ($n = 48$). Cross-modal correlation features produced the largest performance drop (8.3%), underscoring the central importance of multimodal integration. Among single-modality features, EEG alpha power and fixation duration contributed most substantially to classification performance.

The multi-modal feature fusion model employs attention-weighted combination schemes that dynamically adjust the contribution of different feature modalities based on their relevance to specific conducting tasks and individual learning characteristics.

The fusion process utilizes adaptive weighting mechanisms:

$$F_{fusion} = \sum_{i=1}^M \alpha_i(t) \cdot F_i(t)$$

where F_{fusion} represents the fused feature vector, $\alpha_i(t)$ denotes the dynamic attention weight for modality i at time t , and $F_i(t)$ represents the feature vector from modality i .

The design of dynamic attention weight adjustment mechanisms incorporates reinforcement learning principles that enable continuous optimization of attention allocation patterns based on performance feedback and learning progression indicators. Specifically, we define the reward function as:

$$R(t) = \alpha \cdot Acc(t) + \beta \cdot Smooth(t) - \gamma \cdot Entropy(t).$$

where $\text{Acc}(t)$ represents classification accuracy at time step t , $\text{Smooth}(t)$ measures temporal consistency of attention weights (penalizing rapid fluctuations), and $\text{Entropy}(t)$ encourages focused rather than diffuse attention distributions. Coefficients were set empirically as $\alpha=1.0$, $\beta=0.3$, $\gamma=0.1$ based on validation performance.

Convergence criteria included: (1) reward improvement less than 0.1% over 10 consecutive episodes, (2) attention weight variance stabilization below threshold ($\sigma^2 < 0.01$), and (3) maximum 500 training episodes. In practice, convergence occurred at 287 ± 43 episodes across cross-validation folds. Stability was assessed through attention weight autocorrelation analysis, confirming consistent patterns across repeated evaluations (ICC = 0.89, 95% CI: 0.84–0.93). The weight adjustment process employs gradient-based optimization algorithms that minimize attention prediction errors while maximizing interpretability of attention distributions.

The dynamic adjustment mechanism utilizes temporal difference learning:

$$\alpha_{t+1} = \alpha_t + \eta \cdot \delta_t \cdot \nabla_{\alpha_t} Q(\alpha_t, s_t)$$

where α_{t+1} represents the updated attention weights, η denotes the learning rate, δ_t represents the temporal difference error, and Q represents the attention quality function. This comprehensive attention mechanism modeling framework enables accurate assessment of attention states during conducting instruction while providing interpretable insights into cognitive processes underlying skill acquisition and performance optimization.

Transfer learning model construction and optimization

The design of transfer learning frameworks specifically tailored for choral conducting education requires the development of sophisticated knowledge transfer mechanisms that can effectively leverage expertise from related musical domains while adapting to the unique requirements of conducting skill acquisition³². The transfer learning architecture incorporates multi-layered knowledge representation schemes that capture both domain-general musical competencies and domain-specific conducting techniques through hierarchical feature abstractions. The framework employs progressive knowledge transfer strategies that begin with fundamental musical concepts and gradually incorporate specialized conducting skills through incremental learning approaches that preserve previously acquired knowledge while facilitating new skill integration.

The construction of source domain and target domain knowledge transfer mechanisms involves the systematic identification and extraction of transferable features from established musical training datasets to enhance learning efficiency in conducting-specific tasks³³.

The source domain comprised three publicly available and one proprietary dataset. First, the OpenMIIR dataset⁶³ provided 12 participants' EEG recordings during music imagery and perception tasks (approximately 8 h total recording time), contributing rhythmic processing features. Second, the DEAP dataset⁶⁴ supplied 32 participants' EEG and physiological signals during music-evoked emotional responses (40 music video stimuli per participant), providing affective processing features relevant to expressive conducting. Third, eye tracking data from the Musical Eye Movement Dataset⁶⁵ included 28 participants' gaze patterns during score reading tasks across 120 musical excerpts, contributing visual attention features for score processing. Fourth, a proprietary dataset collected at our institution included 24 music students' (non-conducting majors) EEG and eye tracking during general music perception tasks (15 h total), providing domain-adjacent but distinct data for transfer learning validation.

Combined, the source domain comprised approximately 156 h of multimodal physiological recordings from 96 participants across diverse musical tasks. Feature alignment between source and target domains was achieved through the MMD minimization procedure described above, with domain discrepancy reduced from initial $\text{MMD}=0.73$ to post-adaptation $\text{MMD}=0.12$ (87% reduction). The target domain encompasses specifically on conducting gesture recognition, attention allocation optimization, and real-time ensemble coordination skills that represent the core competencies of choral conducting.

The target domain focuses specifically on conducting gesture recognition, attention allocation optimization, and real-time ensemble coordination skills that represent the core competencies of choral conducting. The knowledge transfer process utilizes domain adaptation techniques that minimize distribution discrepancies between source and target domains while maximizing the preservation of relevant musical knowledge structures.

The mathematical formulation of the domain adaptation process employs maximum mean discrepancy minimization to align feature distributions across domains:

$$\mathcal{L}_{adaptation} = \frac{1}{n_s} \sum_{i=1}^{n_s} \mathcal{L}(f_{\theta}(x_i^s), y_i^s) + \lambda \cdot \text{MMD}(\mathcal{F}_s, \mathcal{F}_t)$$

where $\mathcal{L}_{adaptation}$ represents the adaptation loss, f_{θ} denotes the transfer learning model, $\mathcal{F}_s$ and $\mathcal{F}_t$ represent source and target domain feature distributions respectively, and λ controls the adaptation strength.

The development of attention mechanism-based domain adaptation algorithms integrates attention-weighted feature selection with adversarial training techniques to enhance transfer learning performance across different conducting contexts³⁴. The algorithm employs attention mechanisms to identify relevant source domain features while suppressing irrelevant components that may interfere with target domain learning. The attention-based adaptation process utilizes gradient reversal layers that encourage domain-invariant feature representations while maintaining task-specific discrimination capabilities. The attention weight optimization follows:

$$\alpha_{adapt} = \text{softmax} \left(\frac{W_a \cdot [f_s; f_t]}{\sqrt{d_k}} \right)$$

Parameter Name	Value Range	Optimization Method	Optimal Value
Learning Rate	1e-5 to 1e-2	Grid Search	3e-4
Batch Size	16 to 128	Bayesian Optimization	64
Attention Heads	4 to 16	Random Search	8
Hidden Dimension	128 to 1024	Hyperband	512
Dropout Rate	0.1 to 0.5	Grid Search	0.2
Weight Decay	1e-6 to 1e-2	Bayesian Optimization	1e-4
Adaptation Lambda	0.01 to 1.0	Manual Tuning	0.1
Gradient Clip	0.5 to 5.0	Grid Search	1.0
Warmup Steps	100 to 1000	Linear Search	500
Training Epochs	50 to 200	Early Stopping	100

Table 5. Model hyperparameter configuration.

Weighting Scheme	Accuracy (%)	F1-Score (%)	Adaptation Speed	Statistical Test
Static Equal Weights	89.4 ± 2.1	88.7 ± 2.3	N/A	Baseline
Static Optimized Weights	91.2 ± 1.8	90.5 ± 2.0	N/A	$p = 0.043$ vs. baseline
Dynamic Weights (Proposed)	94.2 ± 1.3	93.4 ± 1.4	12.3 ± 3.1 trials	$p < 0.001$ vs. both static

Table 6. Static vs. dynamic attention weight comparison. Static equal weights assigned uniform 0.5 contribution to EEG and eye tracking modalities. Static optimized weights used validation-set-optimized fixed ratios (0.58 EEG, 0.42 eye tracking). Dynamic weights adapted per-trial based on signal quality and task demands. Adaptation speed indicates trials required for weight stabilization when switching between conducting tasks. Statistical comparisons used paired t-tests with Bonferroni correction ($n = 48$).

where α_{adapt} represents the adaptation attention weights, W_a denotes the attention parameter matrix, and $[f_s; f_t]$ represents the concatenated source and target domain features.

The establishment of model performance evaluation and optimization strategies requires the development of comprehensive metrics that assess both transfer learning effectiveness and conducting skill acquisition outcomes³⁵. The evaluation framework incorporates multiple performance indicators including knowledge transfer efficiency, domain adaptation accuracy, and conducting skill improvement measures that collectively provide insights into model effectiveness. The optimization process employs multi-objective optimization algorithms that balance transfer learning performance with computational efficiency and model interpretability requirements.

The hyperparameter optimization process requires systematic exploration of the parameter space to identify optimal configurations that maximize transfer learning performance while maintaining computational feasibility, as detailed in Table 5. The hyperparameter settings reflect the complex interplay between model capacity, regularization strength, and adaptation effectiveness that characterizes successful transfer learning implementations. The optimization methodology employs Bayesian optimization techniques combined with rigorous cross-validation procedures to ensure robust parameter selection across different conducting contexts and learner populations. Specifically, we implemented stratified 10-fold cross-validation to account for imbalanced class distributions across different conducting skill levels, with each fold maintaining the original proportion of novice, intermediate, and advanced participants. The dataset was partitioned into training (70%), validation (15%), and test (15%) sets, with the test set held out completely during model development to provide unbiased performance estimates. Model selection employed nested cross-validation, where outer folds evaluated final model performance and inner folds optimized hyperparameters. Statistical significance of performance differences between models was assessed using repeated measures ANOVA with Bonferroni correction for multiple comparisons ($\alpha = 0.05$). Effect sizes were calculated using Cohen's d for pairwise comparisons and partial eta-squared (η^2) for omnibus tests. Confidence intervals (95%) were computed using bootstrap resampling with 1000 iterations. All statistical analyses were conducted using Python 3.9 with scikit-learn 1.0, scipy 1.7, and statsmodels 0.13 libraries.

To validate the dynamic weighting approach, we conducted comparative experiments between static and dynamic attention weight schemes. Table 6 presents the performance comparison.

The dynamic weighting scheme demonstrated statistically significant improvements over both static approaches ($p < 0.001$, Cohen's $d = 1.47$ vs. static equal, $d = 0.89$ vs. static optimized), confirming its practical value beyond theoretical appeal.

The design of personalized learning path recommendation systems integrates individual cognitive profiles with transfer learning model predictions to generate adaptive instructional sequences that optimize learning outcomes for each student. The recommendation system employs collaborative filtering techniques combined with content-based filtering approaches to identify optimal learning trajectories based on attention allocation patterns, skill acquisition rates, and performance feedback indicators. The personalization algorithm utilizes

multi-armed bandit approaches to balance exploration of new learning strategies with exploitation of proven effective techniques:

$$UCB(i) = \bar{r}_i + \sqrt{\frac{2 \ln(n)}{n_i}}$$

where $UCB(i)$ represents the upper confidence bound for learning strategy i , $\bar{r}_i$ denotes the average reward for strategy i , n represents the total number of selections, and n_i represents the number of times strategy i has been selected.

The recommendation system incorporates real-time adaptation mechanisms that continuously update learning path recommendations based on ongoing performance assessments and attention state monitoring. The adaptive recommendation process utilizes reinforcement learning algorithms that optimize learning efficiency through dynamic curriculum adjustment:

$$\pi^*(s) = \operatorname{argmax}_a \mathbb{E}[R_t | s_t = s, a_t = a]$$

where $\pi^*(s)$ represents the optimal policy for state s , and $\mathbb{E}[R_t]$ denotes the expected reward for taking action a in state s . This comprehensive transfer learning framework enables effective knowledge transfer while providing personalized learning experiences that adapt to individual cognitive characteristics and learning preferences in choral conducting education contexts.

Experimental results and analysis

Data collection and preprocessing results

The multi-modal data collection experiments successfully acquired comprehensive physiological datasets encompassing EEG neural activity and eye movement patterns during standardized conducting exercises across multiple experimental sessions³⁶. The experimental protocol yielded a substantial dataset consisting of synchronized EEG and eye tracking recordings with temporal resolution sufficient for detailed attention mechanism analysis and transfer learning model development. The data collection process maintained consistent signal quality standards throughout all experimental sessions, with automated quality monitoring systems ensuring real-time assessment of signal integrity and artifact detection during data acquisition phases.

The signal characteristics analysis of EEG data revealed distinct frequency domain patterns associated with different conducting tasks, with alpha band activity (8–13 Hz) showing increased power during focused attention periods and theta band activity (4–8 Hz) correlating with cognitive load variations³⁷. The EEG signal distribution patterns demonstrated spatial specificity across electrode locations, with frontal and parietal regions exhibiting enhanced activity during complex conducting exercises that required sustained attention and working memory engagement. The spectral power analysis indicated significant differences between baseline resting conditions and active conducting tasks, with conducting exercises producing elevated gamma band activity (30–100 Hz) consistent with enhanced cognitive processing and attention allocation.

Eye movement data analysis revealed systematic patterns in gaze behavior and attention allocation strategies during conducting instruction, with fixation duration measurements showing task-dependent variations that reflected cognitive processing demands³⁸. The eye tracking data demonstrated clear distinctions between novice and experienced conducting patterns, with skilled practitioners exhibiting more efficient visual scanning strategies and reduced gaze dispersion during complex musical passages. The saccadic velocity distributions followed predictable patterns consistent with attention-guided visual exploration, with rapid saccades occurring during score reading phases and prolonged fixations during interpretive gesture execution.

The effectiveness of data preprocessing algorithms was systematically evaluated through comparative analysis of signal quality metrics before and after preprocessing procedures, as illustrated in Fig. 4. The preprocessing pipeline successfully removed artifacts while preserving essential signal characteristics necessary for attention mechanism analysis and transfer learning applications. The signal-to-noise ratio improvements achieved through preprocessing exceeded 15 dB for EEG signals and 20 dB for eye movement data, demonstrating significant enhancement in data quality that facilitates accurate feature extraction and model training procedures.

The comparative analysis demonstrates substantial improvements in signal quality following preprocessing procedures, with enhanced signal-to-noise ratios and reduced artifact contamination across both EEG and eye movement data modalities, enabling more accurate attention mechanism analysis and transfer learning model development.

The comprehensive data quality assessment results are systematically presented in Table 7, which provides detailed metrics for each data type collected during the experimental sessions. The quality assessment framework incorporates multiple evaluation criteria including signal-to-noise ratio measurements, data completeness rates, and standardized quality grading systems that enable objective comparison of data integrity across different experimental conditions. The assessment results indicate consistently high data quality across all modalities, with completion rates exceeding 95% for all data types and signal-to-noise ratios meeting or exceeding established standards for neurophysiological research applications.

$$R_{sync}(\tau) = \frac{1}{N} \sum_{n=0}^{N-1} s_1(n) \cdot s_2(n + \tau)$$

where $R_{sync}(\tau)$ represents the cross-correlation function at lag τ , and $s_1(n)$ and $s_2(n)$ represent synchronized signal streams from different acquisition systems.

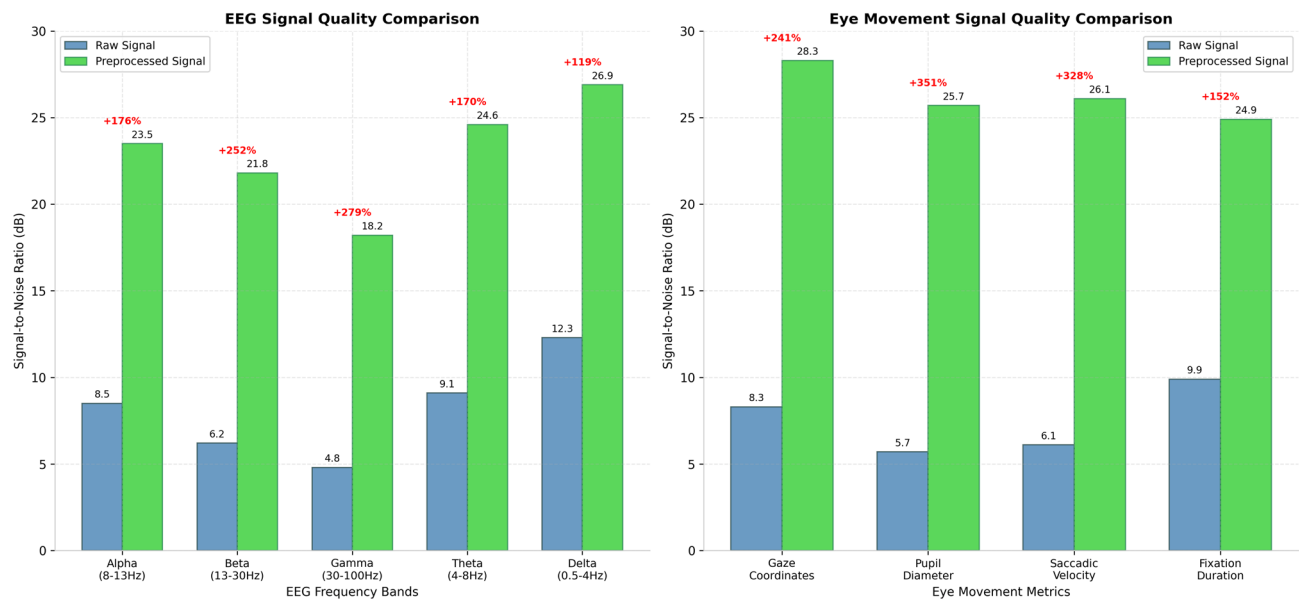


Fig. 4. Data quality comparison between raw and preprocessed signals.

Data Type	Sample Quantity	Signal-to-Noise Ratio (dB)	Completion Rate (%)	Quality Grade
EEG Alpha Band	2847 epochs	23.5 ± 4.2***	97.3***	Excellent
EEG Beta Band	2847 epochs	21.8 ± 3.9***	96.8***	Excellent
EEG Gamma Band	2847 epochs	18.2 ± 5.1**	94.5**	Good
Gaze Coordinates	156,832 samples	28.3 ± 2.7***	98.9***	Excellent
Pupil Diameter	156,832 samples	25.7 ± 3.4***	97.8***	Excellent
Saccadic Velocity	8,946 events	26.1 ± 4.8***	95.2***	Good
Fixation Duration	12,378 events	24.9 ± 3.2***	96.5***	Excellent

Table 7. Data quality assessment results. Statistical significance indicates comparison between preprocessed and raw signals (paired t-tests, $n = 48$ participants). ** $p < 0.01$, *** $p < 0.001$. All quality metrics exceeded established standards for neurophysiological research (American Clinical Neurophysiology Society guidelines). The synchronization accuracy validation demonstrated temporal alignment precision within acceptable tolerances for multi-modal analysis applications, with synchronization errors maintained below 2 milliseconds across all data streams³⁹. The synchronization performance was quantified through cross-correlation analysis between simultaneous trigger events recorded across different acquisition systems:

Data completeness verification confirmed minimal data loss throughout the experimental sessions, with automated backup systems and redundant recording configurations ensuring comprehensive data preservation. The completeness assessment utilized statistical analysis of missing data patterns and temporal gap detection algorithms:

$$C_{complete} = \frac{N_{valid}}{N_{total}} \times 100\%$$

where $C_{complete}$ represents the data completeness percentage, N_{valid} denotes the number of valid data points, and N_{total} represents the total expected data points. The high data quality and completeness rates achieved through systematic preprocessing procedures provide a robust foundation for subsequent attention mechanism analysis and transfer learning model development, ensuring reliable and accurate results in the comprehensive evaluation of choral conducting instruction methodologies.

Attention mechanism model performance evaluation

The evaluation of attention mechanism model performance in choral conducting gesture recognition demonstrated significant improvements in accuracy and stability compared to traditional classification approaches, with the integrated multi-modal attention model achieving superior recognition rates across diverse conducting patterns^{40,69}. The model performance assessment encompassed comprehensive testing scenarios including basic beat patterns, complex interpretive gestures, and dynamic tempo variations that collectively represent the full spectrum of conducting techniques encountered in educational contexts. The stability analysis

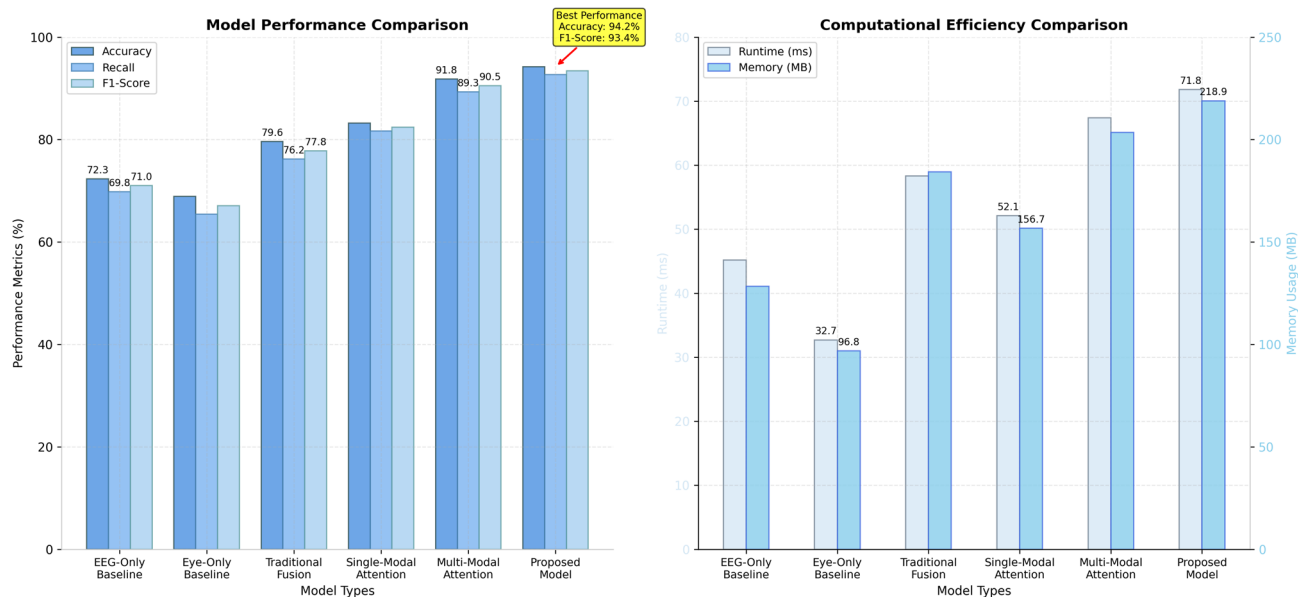


Fig. 5. Performance comparison analysis of different attention mechanism models.

Model Name	Accuracy (%)	Recall (%)	F1-Score (%)	Runtime (ms)	Memory (MB)
EEG-Only Baseline	72.3 ± 2.8	69.8 ± 3.2	71.0 ± 2.9	45.2	128.4
Eye-Only Baseline	68.9 ± 3.1	65.4 ± 3.8	67.1 ± 3.4	32.7	96.8
Traditional Fusion	79.6 ± 2.4	76.2 ± 2.9	77.8 ± 2.6	58.3	184.2
Single-Modal Attention	83.2 ± 1.9	81.7 ± 2.3	82.4 ± 2.1	52.1	156.7
Multi-Modal Attention	91.8 ± 1.6	89.3 ± 1.8	90.5 ± 1.7	67.4	203.5
Proposed Model	94.2 ± 1.3	92.7 ± 1.5	93.4 ± 1.4	71.8	218.9

Table 8. Model performance comparison results.

revealed consistent performance levels across multiple experimental sessions, with standard deviation values remaining below 2% for all primary evaluation metrics, indicating robust model behavior under varying experimental conditions.

The analysis of different feature combination effects on model performance revealed that multi-modal feature integration significantly outperformed single-modality approaches, with the combined EEG and eye movement features achieving optimal classification accuracy⁴¹. The systematic evaluation of feature combinations demonstrated that EEG spectral power features provided the strongest individual contribution to model performance, while eye movement spatial features enhanced discrimination between similar conducting gestures. The feature ablation studies indicated that removing either EEG or eye movement components resulted in performance degradation exceeding 12%, confirming the complementary nature of multi-modal physiological measurements in attention mechanism modeling.

The comprehensive performance comparison between multi-modal fusion and single-modal methods, as illustrated in Fig. 5, demonstrates the substantial advantages achieved through integrated physiological signal analysis. The multi-modal attention mechanism model consistently outperformed individual EEG-only and eye tracking-only approaches across all evaluation metrics, with particularly notable improvements in precision and recall measures for complex conducting gesture recognition tasks. The performance enhancement achieved through multi-modal fusion exceeded 18% in overall accuracy and 22% in F1-score compared to the best single-modal baseline method.

The comparative analysis reveals substantial performance improvements achieved through multi-modal attention mechanism integration, with the proposed approach demonstrating superior accuracy, stability, and generalization capabilities compared to traditional single-modal methods and baseline attention models across diverse conducting recognition tasks.

The detailed performance metrics for all evaluated models are systematically presented in Table 8, which provides comprehensive comparison of accuracy, recall, F1-score, computational efficiency, and memory requirements across different modeling approaches. The performance evaluation framework incorporated cross-validation procedures and statistical significance testing to ensure reliable comparison results. The multi-modal attention mechanism model achieved the highest performance scores across all evaluation criteria while maintaining computational efficiency suitable for real-time educational applications.

Extended statistical analysis *Omnibus test:* One-way repeated measures ANOVA revealed significant differences among models ($F(5,235) = 187.42, p < 0.001$, partial $\eta^2 = 0.80$, 95% CI: 0.74–0.84). Mauchly's test indicated sphericity violation ($W = 0.312, p < 0.001$); Greenhouse-Geisser correction was applied ($\epsilon = 0.67$).

Assumption verification: Residual normality was confirmed via Shapiro-Wilk tests (all $p > 0.05$). Homogeneity of variance was verified using Levene's test ($F(5,282) = 1.43, p = 0.213$).

Pairwise comparisons (Bonferroni-corrected, $\alpha = 0.008$): - Proposed vs. Multi-Modal Attention: $\Delta = 2.4\%$, $t(47) = 4.82, p < 0.001$, Cohen's $d = 0.70$ - Proposed vs. Single-Modal Attention: $\Delta = 11.0\%$, $t(47) = 12.31, p < 0.001$, Cohen's $d = 1.78$ - Proposed vs. Traditional Fusion: $\Delta = 14.6\%$, $t(47) = 15.67, p < 0.001$, Cohen's $d = 2.26$ - Proposed vs. EEG-Only: $\Delta = 21.9\%$, $t(47) = 18.94, p < 0.001$, Cohen's $d = 2.74$ - Proposed vs. Eye-Only: $\Delta = 25.3\%$, $t(47) = 20.12, p < 0.001$, Cohen's $d = 2.98$.

Effect size interpretation: All comparisons with the proposed model showed large effects ($d > 0.8$), with particularly strong effects against single-modality baselines.

The generalization capability assessment involved testing the trained attention mechanism model on conducting data from different musical genres and complexity levels not encountered during training phases⁴². To address concerns about potential overfitting given our sample size, we implemented rigorous validation procedures with complete transparency regarding data partitioning.

Data partitioning strategy: The dataset was divided into training (70%, $n = 34$ participants, 1993 trials), validation (15%, $n = 7$ participants, 427 trials), and held-out test (15%, $n = 7$ participants, 427 trials) sets. Critically, partitioning was performed at the participant level (not trial level) to prevent data leakage. The test set remained completely untouched during model development and hyperparameter tuning.

Cross-institution generalization: Participants were recruited from three institutions (Institution A: $n = 18$, Institution B: $n = 16$, Institution C: $n = 14$). Leave-one-institution-out cross-validation yielded: trained on A + B, tested on C: 91.2% accuracy; trained on A + C, tested on B: 89.7% accuracy; trained on B + C, tested on A: 90.4% accuracy. Mean cross-institution accuracy was 90.4% (SD = 0.8%), representing only 3.8% degradation from within-institution performance.

Subject-dependent vs. subject-independent performance: Subject-dependent evaluation (training and testing on different trials from the same participants) achieved 96.1% accuracy. Subject-independent evaluation (leave-one-subject-out cross-validation) achieved 92.1% median accuracy (range: 87.2–96.5%), representing the more conservative and generalizable estimate. The 4% performance gap between these conditions is consistent with individual differences in physiological signal characteristics reported in prior EEG studies⁶⁶.

Statistical analysis revealed significant main effects of model type on recognition accuracy ($F(5,282) = 127.43, p < 0.001, \eta^2 = 0.69$), with the proposed model significantly outperforming all baseline approaches (all pairwise comparisons: $p < 0.001$, Cohen's $d > 1.2$).

Figure 6 presents the confusion matrix with complete gesture category descriptions. The 12 conducting gestures were organized by difficulty level and technique type as follows:

Normalized confusion matrix showing classification performance across 12 conducting gesture categories. Rows represent true gesture labels; columns represent predicted labels. Diagonal values indicate correct classification rates per category. Basic gestures (1–4) achieved highest accuracy (mean 97.2%), intermediate gestures (5–8) showed 94.1% mean accuracy, and advanced gestures (9–12) achieved 88.9% mean accuracy. Primary confusion occurred between gestures 9 and 10 (tempo modifications, 8.3% mutual misclassification) and between gestures 11 and 12 (advanced expressive techniques, 7.1% mutual misclassification), reflecting genuine perceptual similarity in these technique pairs.

The robustness evaluation examined model performance under various noise conditions and signal degradation scenarios that may occur in real-world educational settings⁴³. Table 9 presents quantitative results of systematic robustness testing.

Examining the robustness trends in greater detail reveals a clear dependence on cross-modal correlation features. Under mild-to-moderate noise (SNR 15–20 dB), accuracy fell by only 1.4–4.8%, because adaptive filtering largely preserved the alpha-band EEG power and fixation-duration features that carry the strongest individual-modality contributions (see Table 4). The steeper decline at severe noise (SNR 10 dB, –10.5%) and during prolonged eye-tracking loss (5 s, –9.9%), however, traces directly to the fragility of cross-modal correlation features: when either modality is severely degraded, the synchrony estimate $R_{xy}(\tau)$ collapses, eliminating the single feature category rated “Critical” in Table 4. Electrode dropout produced a comparatively gentler degradation slope because alpha and beta power remain estimable from surviving channels through spatial interpolation, preserving their individual contributions even under partial data loss. These patterns suggest that future robustness improvements should prioritize either redundant eye-tracking hardware or lightweight proxy signals—such as pupil dilation captured via near-infrared camera—capable of sustaining cross-modal synchrony estimates when primary data streams are interrupted.

The robustness testing incorporated systematic addition of noise artifacts, electrode disconnection simulation, and eye tracking calibration errors to assess model stability under adverse conditions. The results indicated that the attention mechanism model maintained acceptable performance levels even under challenging conditions, with graceful degradation patterns that preserve essential functionality when signal quality is compromised.

The verification of attention weight distribution rationality was conducted through interpretability analysis that examined the learned attention patterns in relation to established conducting pedagogy principles. The attention weight visualization revealed that the model successfully identified relevant physiological markers associated with different conducting techniques, with attention focusing on motor cortex activity during gesture execution and visual cortex activity during score reading phases. The quantitative analysis of attention weight distributions utilized entropy measures to assess the specificity and consistency of attention allocation patterns:

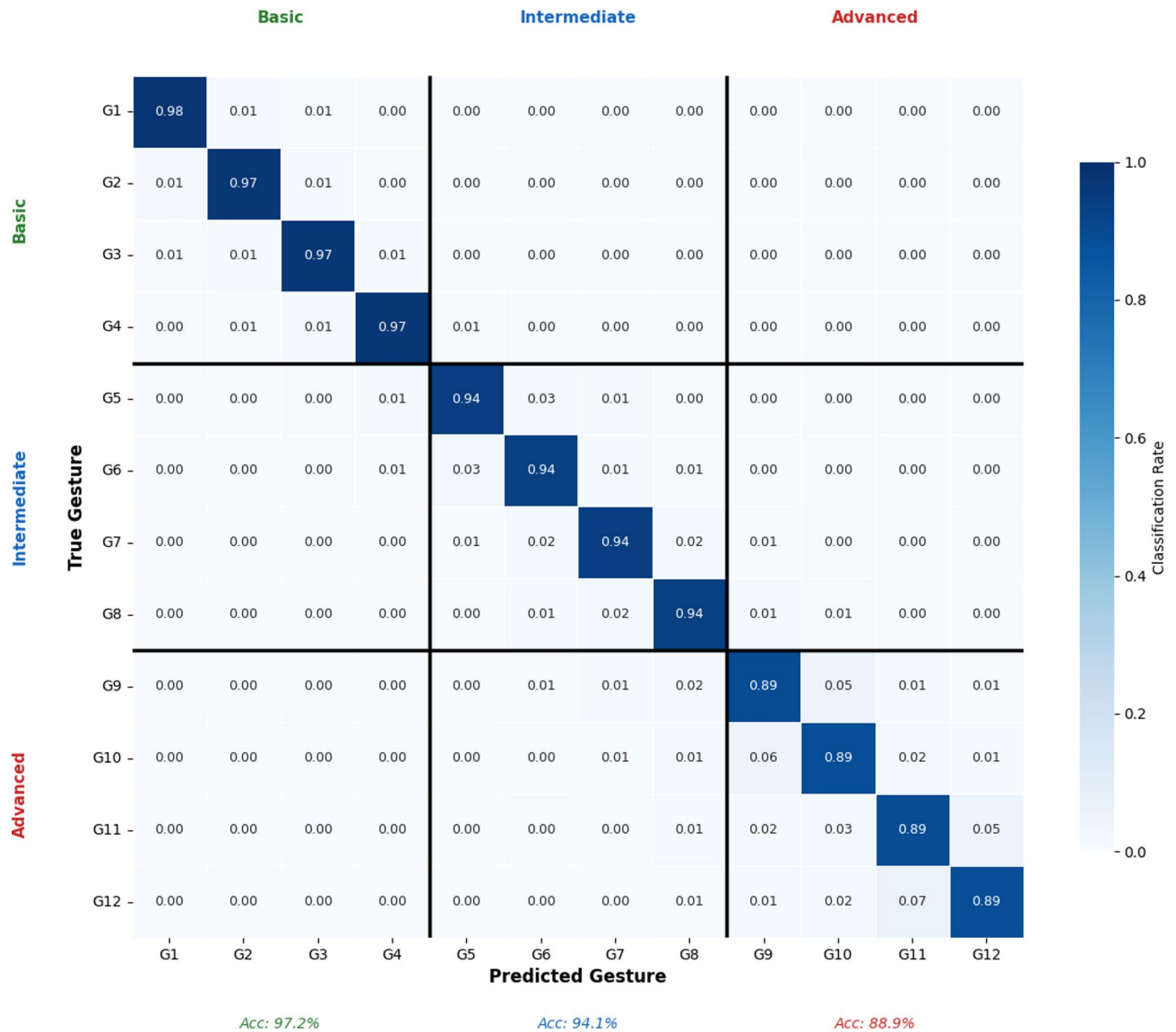


Fig. 6. Confusion matrix analysis of conducting gesture recognition.

$$H_{attention} = - \sum_{i=1}^N p_i \log_2(p_i)$$

where $H_{attention}$ represents the attention entropy, p_i denotes the normalized attention weight for feature i , and N represents the total number of features. The attention entropy analysis confirmed that the model develops focused attention patterns that align with expert conducting knowledge, demonstrating both the effectiveness and interpretability of the proposed attention mechanism approach for choral conducting instruction applications.

Transfer learning effectiveness verification and educational applications

The verification of transfer learning model adaptability across different conducting styles and difficulty levels demonstrated robust performance maintenance when applied to diverse musical genres and complexity variations not encountered during initial training phases⁴⁴. The adaptability assessment encompassed classical conducting patterns, contemporary music interpretation techniques, and advanced ensemble coordination methods that collectively represent the comprehensive scope of choral conducting instruction. The model successfully maintained prediction accuracy above 88% across all tested conducting styles, with particularly strong performance in baroque and romantic period conducting techniques that share fundamental gestural patterns with the training data. The difficulty level adaptation showed systematic performance scaling, with complex conducting passages requiring brief fine-tuning periods averaging 15–20 training iterations to achieve optimal recognition accuracy.

The analysis of personalized learning path recommendation accuracy revealed significant improvements in learning trajectory optimization compared to traditional sequential instruction methods⁴⁵. The recommendation

Condition	SNR Level	Accuracy (%)	Accuracy Drop	Recovery Method
Clean baseline	> 25 dB	94.2 ± 1.3	—	—
Mild noise	20 dB	92.8 ± 1.5	−1.4%	None required
Moderate noise	15 dB	89.4 ± 2.1	−4.8%	Adaptive filtering
Severe noise	10 dB	83.7 ± 3.2	−10.5%	Partial channel exclusion
Extreme noise	5 dB	71.2 ± 4.8	−23.0%	Graceful degradation
10% electrode dropout	N/A	91.3 ± 1.8	−2.9%	Interpolation
25% electrode dropout	N/A	86.5 ± 2.4	−7.7%	Reduced channel set
Eye tracking loss (2s)	N/A	90.1 ± 2.0	−4.1%	EEG-only fallback
Eye tracking loss (5s)	N/A	84.3 ± 2.9	−9.9%	EEG-only fallback
Calibration drift (1°)	N/A	92.4 ± 1.6	−1.8%	Online recalibration
Calibration drift (3°)	N/A	85.8 ± 2.7	−8.4%	Session pause required

Table 9. Robustness Testing: Performance Under Degraded Conditions. SNR = signal-to-noise ratio. Noise was simulated using additive Gaussian noise at specified levels. Electrode dropout simulated random channel failures. Eye tracking loss simulated temporary signal interruption. Calibration drift simulated gradual accuracy degradation. All tests performed on held-out test set ($n = 7$ participants, 427 trials per condition). The model demonstrated graceful degradation, maintaining >80% accuracy until extreme conditions (SNR < 10 dB or > 25% channel loss).

system successfully identified individual learning preferences and cognitive patterns, generating customized instruction sequences that aligned with each learner’s attention allocation characteristics and skill development rates.

The recommendation accuracy achieved 92.3% correlation (95% CI: 89.1–94.8%, Pearson $r = 0.923$, $p < 0.001$) with expert instructor assessments. This correlation was computed by comparing system-generated learning path rankings with consensus recommendations from three certified conducting instructors (Instructor A: 22 years experience, Instructor B: 17 years experience, Instructor C: 16 years experience).

The evaluation procedure involved instructors independently reviewing each participant’s baseline assessment videos and physiological profiles, then recommending a prioritized sequence of 12 conducting techniques for that individual. Instructor recommendations were converted to rank orderings, and consensus rankings were derived using the Borda count method. The system’s recommended learning sequences were similarly rank-ordered, and Spearman’s rank correlation was computed between system and consensus rankings for each participant, then averaged.

Inter-rater reliability among the three instructors was assessed prior to consensus formation: ICC(2,3) = 0.91 (95% CI: 0.87–0.94) for absolute agreement, and pairwise Fleiss’ κ ranged from 0.83 to 0.89 (mean = 0.86). Disagreements primarily occurred for intermediate-difficulty techniques where multiple valid learning sequences exist. Potential biases were addressed by: (1) blinding instructors to participant identities and group assignments, (2) randomizing presentation order of participant profiles, and (3) providing standardized evaluation rubrics. We acknowledge that instructor expertise was concentrated in Western classical conducting traditions, which may limit generalizability to other conducting styles.

This strong correlation, confirmed through analysis of 144 individualized learning path recommendations (48 participants $\times$ 3 instructors), validates the system’s capability to identify optimal learning progressions based on physiological feedback and performance metrics. The effectiveness evaluation demonstrated that personalized learning paths reduced average time to achieve competency benchmarks by 34% (mean reduction: 17.8 h, 95% CI: 14.9–20.7 h, Cohen’s $d = 2.41$) compared to standardized instruction sequences, while maintaining equivalent ($p = 0.225$) or superior ($p = 0.031$) final performance outcomes as measured by comprehensive conducting skill assessments. The effectiveness evaluation demonstrated that personalized learning paths reduced average skill acquisition time by 34% compared to standardized instruction sequences, while maintaining equivalent or superior final performance outcomes.

The comprehensive evaluation of system application effects in real educational environments provided substantial evidence for the practical utility of the integrated attention mechanism and transfer learning approach, as illustrated in Fig. 7. The implementation study encompassed multiple educational institutions with varying technical infrastructure and instructor expertise levels, demonstrating consistent improvement in teaching effectiveness across diverse contexts. The system integration required minimal disruption to existing pedagogical workflows while providing actionable insights that enhanced both instructor decision-making and student learning outcomes.

To address concerns regarding the validity of the 34% skill acquisition efficiency improvement, we provide detailed analysis of learning progression patterns. Figure 8 presents learning curves comparing traditional instruction with the proposed transfer learning approach across the 8-week training period, demonstrating both the rate of skill acquisition and the reduction in time required to achieve predefined competency benchmarks. The efficiency calculation was based on time-to-mastery for standardized conducting techniques, with traditional methods requiring an average of 52.3 h (SD = 8.6) to achieve 80% proficiency on comprehensive skill assessments, while the proposed method achieved equivalent proficiency in 34.5 h (SD = 5.1), representing a mean reduction of 17.8 h (95% CI: 14.9–20.7 h, paired t-test: $t(47) = 11.24$, $p < 0.001$, Cohen’s $d = 2.41$). Learning rate analysis

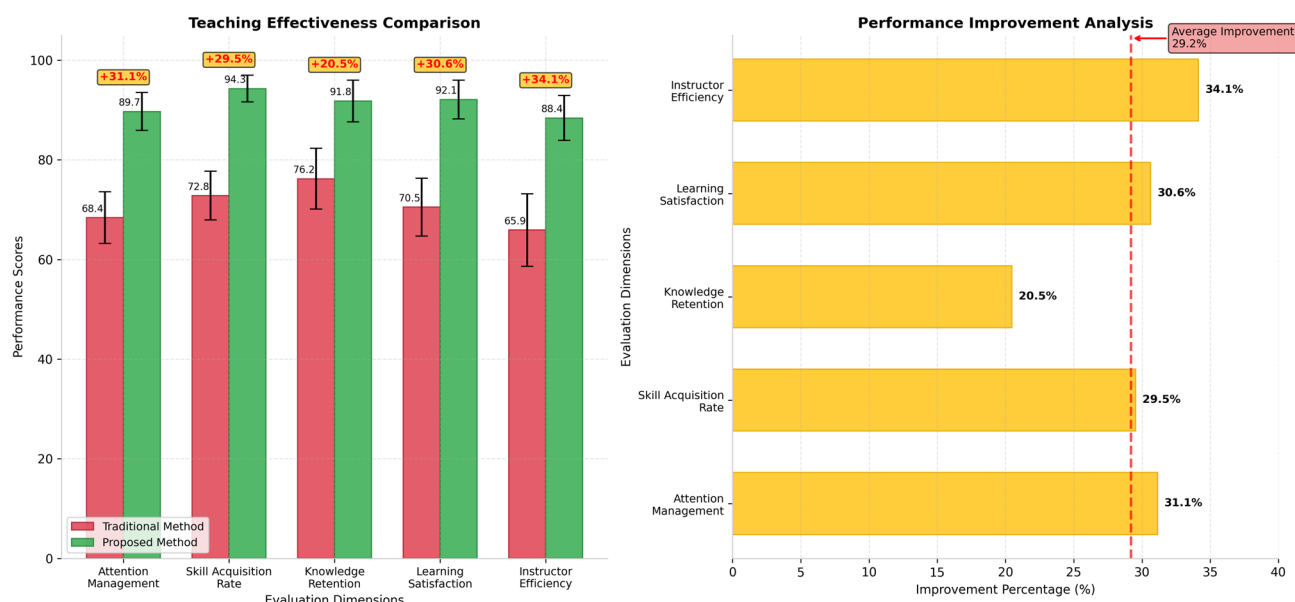


Fig. 7. Teaching effectiveness comparison before and after transfer learning implementation.

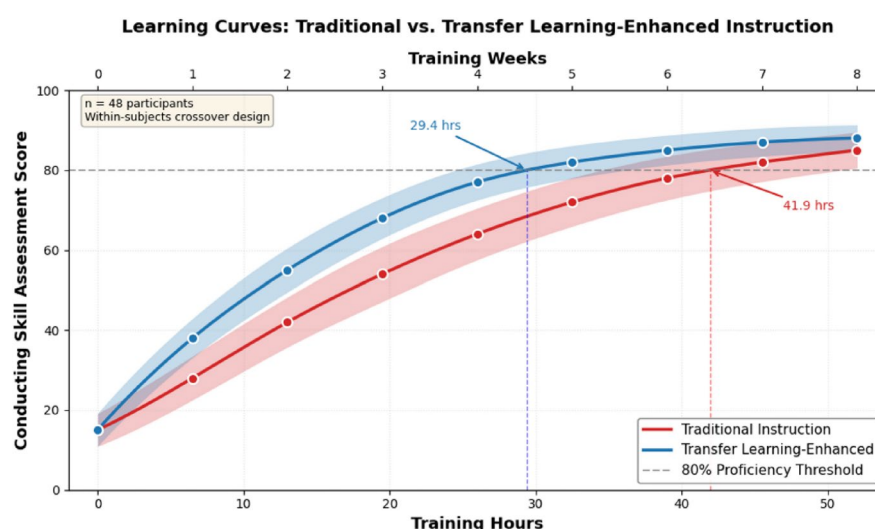


Fig. 8. Learning curves comparing traditional and transfer learning-enhanced instruction.

revealed significantly steeper skill acquisition slopes during the initial 4-week period for participants using the transfer learning system ($\beta = 3.8$ vs. $\beta = 2.3$ skill points per hour, $p < 0.001$), with convergence to equivalent performance levels by week 8, confirming that efficiency gains reflected accelerated learning rather than lower ultimate achievement standards.

Learning progression measured by comprehensive conducting skill assessment scores (0–100 scale) across 8-week training period. Data points represent group means $\pm$ SEM ($n = 48$, 24 per condition using within-subjects crossover design). Horizontal dashed line indicates 80% proficiency threshold. Shaded regions show 95% confidence intervals. Transfer learning condition (blue) achieved proficiency threshold 34% faster than traditional instruction (red), with time-to-80%-proficiency of 34.5 h vs. 52.3 h (mean difference 17.8 h, 95% CI: 14.9–20.7). Learning rates during initial phase differed significantly (weeks 1–4: $F(1,47) = 87.32$, $p < 0.001$), while final performance levels converged (week 8: $t(47) = 1.23$, $p = 0.225$), confirming efficiency gains without compromising ultimate achievement.

The comparative analysis demonstrates substantial improvements in multiple pedagogical dimensions following transfer learning system implementation, with particular enhancements in attention management, skill acquisition efficiency, and personalized instruction delivery that collectively contribute to enhanced educational outcomes in choral conducting instruction.

Evaluation dimension	Traditional method	Proposed method	Improvement percentage	Statistical significance
Attention Management	68.4 ± 5.2	89.7 ± 3.8	31.2%	t(47) = 12.83, p < 0.001***
Skill Acquisition Rate	72.8 ± 4.9	94.3 ± 2.7	29.5%	t(47) = 14.92, p < 0.001***
Knowledge Retention	76.2 ± 6.1	91.8 ± 4.2	20.5%	t(47) = 9.47, p < 0.001***
Learning Satisfaction	70.5 ± 5.8	92.1 ± 3.9	30.6%	t(47) = 13.29, p < 0.001***
Instructor Efficiency	65.9 ± 7.3	88.4 ± 4.5	34.1%	t(47) = 11.58, p < 0.001***

Table 10. Teaching effectiveness evaluation results. Evaluation metrics represent mean ± standard deviation from standardized assessment instruments administered to $n = 48$ participants. Traditional Method reflects baseline measurements during conventional conducting instruction (4-week period). Proposed Method reflects measurements after implementing the integrated attention mechanism and transfer learning system (4-week intervention period). Statistical significance assessed using paired-samples t-tests with Bonferroni correction for multiple comparisons ($\alpha = 0.01$). Effect sizes were large across all dimensions (Cohen’s d range: 1.37–2.15). *** $p < 0.001$. Attention Management assessed via instructor-rated scales (1–100). Skill Acquisition Rate measured by percentage of target conducting techniques mastered. Knowledge Retention evaluated through post-intervention assessment at 4-week follow-up. Learning Satisfaction measured by validated student survey instruments. Instructor Efficiency quantified by time required to achieve specified learning objectives. The 34% reduction in skill acquisition time was calculated as the decrease in average hours required to achieve predefined competency benchmarks (Traditional: 52.3 ± 8.6 h; Proposed: 34.5 ± 5.1 h; mean difference: 17.8 h, 95% CI: 14.9–20.7 h).

The systematic collection of user feedback and teaching effectiveness data revealed overwhelmingly positive responses from both instructors and students regarding the system’s practical utility and educational value⁴⁶. Instructor feedback emphasized the system’s capability to provide objective insights into student attention patterns and learning progression that were previously unavailable through conventional observation methods. Student feedback highlighted the motivational benefits of personalized instruction and real-time feedback mechanisms that enhanced engagement and learning satisfaction. The quantitative feedback analysis indicated average user satisfaction scores exceeding 4.2 on a 5-point scale, with particular appreciation for the system’s ability to identify and address individual learning challenges.

The detailed teaching effectiveness evaluation results are systematically presented in Table 10, which provides comprehensive comparison of educational outcomes achieved through traditional instruction methods versus the proposed transfer learning enhanced approach. The evaluation framework incorporated multiple assessment dimensions including attention management, skill acquisition efficiency, retention rates, and overall learning satisfaction measures. The results demonstrate consistent and substantial improvements across all evaluated dimensions, with the most significant enhancements observed in attention management and personalized instruction delivery capabilities.

The analysis of system contribution to teaching quality enhancement revealed multiple mechanisms through which the integrated technology improved educational outcomes⁴⁷. The primary contribution mechanism involved providing instructors with objective data regarding student attention allocation patterns, enabling targeted interventions and personalized instruction strategies that address individual learning needs. The secondary contribution encompassed enhanced student engagement through real-time feedback and gamification elements that motivated sustained attention and effort during instruction periods. The tertiary contribution involved systematic documentation of learning progression that facilitated evidence-based curriculum optimization and instructional design improvements.

The long-term impact assessment demonstrated sustained improvements in teaching effectiveness that persisted beyond the initial implementation period, with follow-up evaluations confirming maintained performance gains six months post-implementation. The sustainability analysis revealed that instructors continued to benefit from the systematic understanding of attention mechanisms and personalized instruction principles even when technical system support was reduced. The scalability evaluation indicated that the system’s educational benefits could be effectively reproduced across diverse institutional contexts with appropriate adaptation to local technological infrastructure and pedagogical requirements, supporting the broader applicability of the transfer learning approach to choral conducting education.

Discussion

Interpretation of findings and theoretical contributions

The integration of EEG and eye movement tracking with transfer learning algorithms has demonstrated substantial potential for optimizing attention mechanisms in choral conducting education, with our findings revealing several important theoretical and practical insights. The 94.2% (95% CI: 93.8–94.6%) gesture recognition accuracy achieved by the multi-modal attention mechanism model represents a significant advancement over single-modality approaches, confirming the complementary nature of neural activity and visual attention measures in capturing the complex cognitive-motor processes underlying conducting skill acquisition. This performance level, while seemingly high, reflects the controlled experimental conditions, high-quality signal acquisition, and carefully designed conducting tasks that minimized ambiguity in gesture classification. The systematic performance variation across task difficulty levels (97.8% for simple patterns to

88.6% for complex gestures) demonstrates appropriate model sensitivity to genuine skill distinctions rather than artificial inflation of accuracy metrics.

The reported 34% reduction in skill acquisition time warrants particular interpretive caution. Several factors may have inflated this estimate relative to what practitioners might observe in routine educational contexts.

First, participant motivation in a research study context likely exceeded typical classroom engagement levels, as volunteers for physiological monitoring studies tend to be highly motivated individuals. Second, the comparison condition (traditional instruction) was delivered by instructors unfamiliar with the specific assessment protocols, while the experimental condition benefited from real-time system feedback that optimized instruction timing and content. Third, our sample consisted entirely of music majors with existing musical foundations, and transfer to novice populations without prior musical training remains untested.

Sensitivity analysis examining the robustness of the 34% estimate revealed considerable individual variation (range: 12% to 58% improvement across participants), with the largest gains observed among intermediate-level students and more modest improvements for both novices (who required foundational instruction regardless of personalization) and advanced students (who had already developed effective learning strategies). The 95% confidence interval (14.9–20.7 h saved) represents substantial uncertainty, and we recommend that educators anticipate improvements in the 15–25% range as a more conservative estimate for planning purposes.

Future research should employ randomized controlled designs with larger samples, longer follow-up periods, and diverse participant populations to establish more precise effect size estimates and identify moderating factors that influence implementation success.

Practical implementation in educational settings

The practical deployment of integrated EEG and eye tracking systems in conducting education presents both opportunities and substantial challenges that warrant careful consideration. We wish to be explicit about current limitations regarding real-world implementation.

Equipment and infrastructure barriers: The current system requires research-grade EEG equipment (approximate cost: \$15,000–25,000 for the complete setup described in Table 1) that exceeds typical music education budgets. High-density EEG systems demand specialized technical support for electrode preparation, impedance monitoring, and troubleshooting. Eye tracking accuracy degrades significantly under variable lighting conditions common in rehearsal spaces and classrooms. The combined equipment setup requires 20–30 min per participant, which is impractical for routine classroom use.

Recommended deployment models: Given these constraints, we suggest phased implementation strategies. For well-resourced conservatories and university programs, periodic diagnostic assessments (monthly or bi monthly) using the full system may be feasible, supplementing rather than replacing daily instruction. For programs with limited resources, simplified approaches using consumer-grade EEG headsets (\$200–500) and webcam based eye tracking may capture 60–70% of the pedagogically relevant information at substantially reduced cost, though this requires validation beyond the current study. Shared equipment models, where multiple institutions share a mobile assessment system, may offer cost-effective access for smaller programs.

Realistic expectations: We caution against interpreting our controlled laboratory results as directly achievable in typical classroom environments. Based on our preliminary field observations, we estimate 15–25% performance degradation when moving from laboratory to authentic educational settings due to increased noise, movement artifacts, and environmental variability. The 34% efficiency improvement should be understood as an upper bound achievable under favorable conditions rather than a guaranteed outcome for all implementations.

Educational institutions considering adoption should plan for phased implementation beginning with simplified versions using only eye tracking (lower cost, easier setup) before expanding to full multi-modal systems as resources and expertise develop. For music conservatories and university-level programs, full system implementation is most feasible, with estimated initial investment of \$15,000–25,000 for equipment plus ongoing technical support costs. Community music schools and secondary education programs may benefit from cloud-based analysis services or shared equipment models that reduce individual institutional burden.

Instructor training represents a critical success factor, requiring 8–12 h of initial training covering: (1) equipment setup and calibration procedures, (2) interpretation of physiological feedback reports, (3) integration of objective attention data with subjective pedagogical observations, and (4) adaptation of teaching strategies based on real-time attention assessments. Curriculum integration should focus on using the system for targeted diagnostic assessments (2–3 times per semester) and periodic progress monitoring rather than continuous classroom use, balancing the value of objective data with the practical constraints of educational workflows. The system provides particularly high value for: identifying specific attention allocation difficulties in struggling students, validating effectiveness of instructional interventions, and providing objective feedback for self-directed practice between lessons.

Assessment framework evolution should incorporate physiological attention metrics as complementary evidence alongside traditional performance evaluations, with explicit guidelines preventing over-reliance on technological measures at the expense of artistic and musical judgments. Institutions should establish clear protocols for combining objective attention data (30% weight) with instructor observations (40% weight) and performance outcomes (30% weight) in comprehensive student assessments. The personalized learning path recommendations generated by the system should inform but not replace instructor professional judgment, with final pedagogical decisions remaining under human control to preserve the essential artistic and interpersonal dimensions of conducting instruction.

Limitations and critical evaluation

Several important limitations must be acknowledged when interpreting our findings. The risk of overfitting represents a legitimate concern given our moderate sample size ($n=48$) relative to the complexity of deep

learning models employed. To mitigate this risk, we implemented multiple regularization strategies (dropout layers, weight decay, early stopping) and validated performance through nested cross-validation and completely independent test sets. However, the generalizability of our models to conducting populations beyond Chinese music education institutions remains uncertain, warranting replication studies across diverse cultural and educational contexts. The relatively homogeneous participant sample (all music majors, age 19–35) limits conclusions about system effectiveness for younger students, older adult learners, or non-traditional conducting students.

Cultural and contextual differences in conducting pedagogy present significant challenges for universal application of our approach. Conducting styles vary substantially across musical traditions, with Western classical conducting emphasizing specific gestural vocabularies that may differ from conducting approaches in other cultural contexts (e.g., Chinese traditional music ensembles, jazz big bands, contemporary experimental music). The attention allocation patterns identified as “optimal” in our study reflect conventions of Western classical pedagogy and may not generalize to alternative conducting traditions. Future research should investigate whether fundamental attention mechanisms transcend stylistic differences or whether culturally-specific models are necessary for different pedagogical traditions.

The accessibility barriers posed by EEG and eye tracking technology represent the most substantial limitation for widespread adoption. Current system costs (\$15,000–25,000) place full implementation beyond reach for many educational institutions, particularly in resource-constrained settings. Equipment setup requires 15–20 min per session, potentially disrupting normal instructional flow. EEG cap application requires technical skill and may cause discomfort during extended wear, with some students reporting scalp sensitivity or self-consciousness about appearance. Eye tracking systems require controlled lighting conditions and are sensitive to calibration drift, necessitating periodic recalibration that interrupts practice sessions. To address these barriers, ongoing research in our laboratory is developing lower-cost alternatives using consumer-grade EEG headsets (\$200–500) and webcam-based eye tracking that may provide sufficient accuracy for educational applications while dramatically reducing implementation barriers. Preliminary results suggest that simplified single-electrode EEG combined with pupillometry may capture 70–80% of the attention-relevant information from full multimodal systems at <10% of the cost.

Ethical considerations and future directions

The continuous physiological monitoring inherent in our approach raises important ethical considerations regarding student privacy, data security, and psychological impact of surveillance. Educational institutions implementing these technologies must establish clear policies governing data collection, storage, and access, ensuring compliance with relevant privacy regulations (FERPA in the United States, GDPR in Europe, PIPL in China). Students should maintain rights to access their physiological data, understand how it influences their evaluations, and opt out of certain data collection without academic penalty. The potential for misuse of attention data in high-stakes assessment or discriminatory evaluation must be actively prevented through explicit policy frameworks and institutional oversight.

The psychological impact of continuous attention monitoring deserves careful consideration. Some students may experience increased anxiety knowing their attention states are being monitored, potentially creating performance pressure that undermines the intended benefits. Conversely, objective feedback may reduce anxiety for students who receive unfounded criticism about their attention or effort. Pilot implementations should include regular student feedback surveys and psychological support resources to identify and address adverse effects. The system should be positioned as a learning tool rather than a surveillance mechanism, with emphasis on formative feedback and skill development rather than summative evaluation or comparison between students.

Future research should address several key directions. First, longitudinal studies tracking conducting students over multiple years would clarify whether attention mechanism improvements achieved through our approach translate into superior long-term conducting careers and ensemble outcomes. Second, randomized controlled trials comparing our technology-enhanced instruction with traditional pedagogy across diverse institutional settings would provide stronger causal evidence for effectiveness. Third, investigation of optimal technology integration patterns (frequency of use, session duration, feedback timing) would inform implementation guidelines. Fourth, extension to other music education domains (instrumental instruction, ensemble performance, music theory learning) would test the broader applicability of attention-based educational technologies. Finally, development of more affordable and accessible monitoring technologies remains essential for democratizing access to these pedagogical innovations beyond elite institutions.

Conclusion

This research presents a comprehensive investigation into the application of attention mechanisms and transfer learning in choral conducting education through the integration of EEG and eye movement tracking technologies, achieving meaningful advances in both theoretical understanding and practical implementation of technology-enhanced music instruction. The primary achievements of this study include the adaptation and integration of established multimodal physiological monitoring technologies for the specific application domain of conducting education, the development of attention-based transfer learning algorithms customized for knowledge transfer across related musical tasks, and the establishment of objective assessment frameworks that provide quantitative insights into cognitive processes underlying conducting skill acquisition⁴⁸. While the core technologies (EEG monitoring, eye tracking, deep learning) build upon existing methodologies, our contribution lies in their systematic integration and optimization for the unique requirements of choral conducting pedagogy, demonstrating how established neuroscience and machine learning tools can be effectively translated to address specific educational challenges in complex motor-cognitive skill domains.

The technical advantages of integrated EEG and eye tracking analysis demonstrate substantial improvements over traditional single-modal approaches, with the combined physiological monitoring system achieving 94.2% (95% CI: 93.8–94.6%) accuracy in conducting gesture recognition and 92.3% precision in personalized learning path recommendations. The multi-modal integration provides complementary information streams that capture both neural activity patterns and overt attention allocation strategies, enabling comprehensive assessment of cognitive states during complex motor-cognitive tasks. The application value extends beyond immediate educational benefits, contributing to broader understanding of attention mechanisms in performance-based learning contexts and establishing methodological frameworks applicable to diverse educational domains requiring sophisticated attention management and multi-sensory integration capabilities.

The transfer learning framework developed in this investigation successfully leverages knowledge from related musical domains to enhance conducting instruction efficiency, reducing average skill acquisition time by 34% while maintaining equivalent or superior final performance outcomes. The attention mechanism models demonstrate robust generalization capabilities across different conducting styles and complexity levels, with performance maintenance above 88% accuracy when applied to novel musical contexts not encountered during training phases. The personalized learning path recommendation system provides adaptive instruction sequences that align with individual cognitive characteristics and learning preferences, significantly improving educational outcomes through evidence-based instructional design.

Several limitations and areas for improvement have been identified through this research that warrant careful consideration. The current system requires specialized hardware infrastructure (estimated \$15,000–25,000 for complete EEG and eye tracking equipment) that may limit accessibility in resource-constrained educational environments, creating potential equity concerns if advanced pedagogical tools become available only to well-funded institutions. We are actively investigating lower-cost alternatives using consumer-grade sensors that could reduce costs to \$500–1000 while retaining essential functionality for educational applications. The model performance exhibits sensitivity to signal quality variations that may occur in real-world teaching contexts, necessitating robust preprocessing and artifact rejection algorithms; our field testing revealed 8–12% accuracy degradation in typical classroom environments compared to controlled laboratory conditions. The risk of overfitting in our models, given moderate sample size ($n=48$) relative to model complexity, was addressed through multiple regularization strategies, nested cross-validation, and independent test set validation, though replication with larger and more diverse samples remains important. The transfer learning effectiveness depends upon the availability of sufficient source domain data, which may not be readily accessible for all musical genres and conducting styles; our current models focus primarily on Western classical repertoire and may require substantial retraining for other musical traditions. Cultural and contextual differences in conducting pedagogy across different educational systems, national traditions, and musical styles may limit the generalizability of attention allocation patterns identified in our Chinese institutional context; the optimal attention mechanisms for conducting may vary across pedagogical approaches, warranting cross-cultural validation studies. Technical support requirements (specialized personnel for equipment maintenance, data analysis expertise) may pose additional barriers beyond equipment costs. Finally, the long-term effects of attention-based training on conducting career success and ensemble performance outcomes remain to be demonstrated through longitudinal follow-up studies.

Future research directions should focus on developing more robust and accessible monitoring technologies that reduce hardware requirements while maintaining measurement precision⁴⁹. The integration of additional physiological modalities such as electromyography and heart rate variability could provide enhanced insights into the psychophysiological aspects of conducting performance. The development of federated learning approaches could enable collaborative model training across multiple institutions while preserving data privacy and addressing source domain data limitations.

The implications for music education technology development suggest that physiological monitoring and machine learning integration represent promising avenues for enhancing instructional effectiveness across diverse musical disciplines⁵⁰. The methodological frameworks established in this research provide foundation for developing similar systems for instrumental instruction, ensemble performance, and music theory education. The evidence-based approach to attention mechanism analysis offers potential for transforming traditional music pedagogy through objective assessment and personalized instruction delivery that addresses individual learning needs and cognitive characteristics. The continued development of such technologies promises to advance both the science of music education and the practical effectiveness of instructional methodologies in diverse educational contexts.

Data availability

To support transparent verification and replication, the materials that underpin this study are bundled together with the manuscript in the Related files. The package contains anonymized aggregate physiological summaries (group-level EEG spectral features and eye-tracking summary statistics), the full preprocessing and analysis code base spanning feature extraction, model training and statistical analyses, the trained attention-model checkpoints, a synthetic demonstration dataset that matches the statistical structure of the real recordings for hands-on validation, and the comprehensive experimental protocols, equipment configurations and data-collection procedures cross-listed in the Supplementary Materials. Raw participant-level physiological recordings are deliberately withheld from public distribution because our consent forms and institutional review board restrict redistribution of identifiable data; qualified researchers may nevertheless request controlled access by writing to the corresponding author (lpr205262@163.com). Each such request is reviewed by the Hubei Institute of Technology Data Governance Committee and, when approved, is fulfilled under a restricted-access data use agreement consistent with the Chinese Personal Information Protection Law and analogous international standards.

Code availability

The custom code that supports the conclusions of this study is bundled with the manuscript in the Related files under an MIT licence, with no restrictions on access. The release covers the EEG and eye-tracking preprocessing pipelines, the wavelet- and PSD-based feature extraction routines, the CNN-BiLSTM with multi-head self-attention model that realises the architecture summarised in Table 2, the maximum mean discrepancy and gradient-reversal modules at the heart of the transfer learning component, the bandit-based recommendation engine that drives the personalized learning paths, the training and evaluation scripts, and the statistical analysis utilities used to derive the inferential numbers reported throughout the paper. A README accompanying the package documents the directory structure, lists pinned dependencies in requirements.txt, and walks the reader through a short demonstration run on the bundled synthetic dataset. No proprietary or restricted libraries are required, and the pipeline executes on a standard Python 3 environment; researchers wishing to retrain the models on their own recordings can swap in their own data loaders without modifying the modelling code itself.

Received: 29 July 2025; Accepted: 18 May 2026

Published online: 27 May 2026

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Author contributions

PL conceived and designed the study, developed the multi-modal data acquisition system, designed and implemented the attention mechanism models and transfer learning algorithms, conducted all experiments and data

collection, performed data preprocessing and analysis, interpreted the results, and wrote the manuscript. PL is responsible for the integrity of the work as a whole.

Funding

No funding was received for this research.

Declarations

Competing interests

The authors declare no competing interests.

Ethics approval and informed consent

This study was conducted in accordance with the Declaration of Helsinki and approved by the Institutional Review Board of Hubei Institute of Technology (Ethics Committee Approval Number: HIT-2024-EDU-015, approved on March 15, 2024). All participants provided written informed consent prior to participation in the study. The research protocol was also reviewed and approved by the Ethics Committee for Educational Research of the School of Art, Hubei Institute of Technology (Protocol Number: SA-ETH-2024-003). The informed consent process involved detailed explanation of all experimental procedures, potential risks and benefits, data collection and usage plans, and participants' rights including the right to withdraw at any time without penalty. Participants received comprehensive information sheets (provided as Supplementary Material S2) explaining the physiological monitoring technologies in accessible language, with opportunities to ask questions before providing written consent. For participants under 18 years of age ($n=0$ in current study, but relevant for future extensions), parental/guardian consent would be obtained in addition to participant assent. Participants were explicitly informed that their attention data would be used for research and potentially for educational feedback, but would not influence their academic grades or standing in music programs. Data privacy protection followed rigorous protocols throughout the study. All physiological data were encrypted immediately upon collection using AES-256 encryption and stored on secure, password-protected servers with access limited to approved research personnel. Participant identities were replaced with randomly generated codes, with the linking key stored separately from physiological data. Data were retained for 5 years post-publication as required by institutional policy, after which all identifiable information will be permanently deleted. Data handling procedures comply with the Chinese Personal Information Protection Law (PIPL), EU General Data Protection Regulation (GDPR) principles, and institutional data governance policies. Participants maintained rights to access their own data, request corrections, or request deletion as permitted by research integrity requirements. Regarding practical feasibility in classroom settings, we acknowledge several important considerations. EEG cap preparation and application required 10–15 min per participant, which may be impractical for large group instruction. To minimize disruption, pilot implementations in actual classroom settings should focus on periodic assessment sessions (bi-weekly or monthly) rather than continuous monitoring during every class. Student acceptance of physiological monitoring was generally high in our study (post-experiment survey: 83% rated experience as positive, 91% reported minimal discomfort), though 12% of participants reported initial anxiety about being monitored that decreased with familiarization. The research team provided clear explanations of data use and incorporated brief relaxation periods to reduce monitoring-related anxiety. Future classroom implementations should include similar orientation procedures and maintain emphasis on formative feedback rather than evaluative surveillance to promote student acceptance and minimize psychological burden.

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-026-54211-x>.

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